Prague University of Economics and Business
Faculty of Informatics and Statistics

Benford's Law and the Statistical Detection of Election Fraud

BACHELOR THESIS

Study program: Mathematical Methods in Economics
Specialization: Data Analyses and Modeling

Author: Kristýna Bednářová
Supervisor: Mgr. Jana Cibulková, Ph.D.

Prague, May 2025

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Mgr. Jana Cibulková, Ph.D., whose motivating feedback throughout the process was of inestimable value.

Many thanks to my parents, Vítek, friends and coworkers. Your support and kind comments have meant the world to me.

Abstract

Elections worldwide are essential for democratic governance, but election fraud seriously threatens the integrity that underpins democratic societies. This thesis explores the applicability of Benford's Law as a forensic tool for detecting anomalies in electoral data, a possible manifestation of election fraud, by introducing a novel, reproducible methodology, thoughtfully designed by the author through the integration of multiple established approaches. It combines compliance tests of the leading and last digits distributions according to Benford's law and other indicative measures. These include correlation of vote shares, turnout, and the share of valid votes, and the digital development pattern. The author was able to reach similar conclusions as prominent international organisations, such as the OSCE or the EU, by applying the proposed methodology to presidential elections with different political contexts: the Czech Republic (2023), the United States (2024), and Belarus (2020). While the Czech and U.S. elections showed no significant deviations from expected digit distribution patterns, the Belarusian data revealed multiple irregularities, particularly in the first and last digits of vote counts. This illustrates that statistical analysis of digit patterns can serve as a useful preliminary indicator of electoral manipulation.

Keywords

Benford's Law, election fraud, leading digit, electoral integrity

Abstrakt

Volby na celém světě mají zásadní význam jako mechanismus demokracie, ale volební podvody vážně ohrožují volební integritu, na které volby stojí. Tato práce zkoumá jak se dá Benfordův zákon použít jako forenzní nástroj pro odhalování anomálií ve volebních výsledcích, které jsou možným projevem volebních podvodů, a to zavedením nové, reprodukovatelné metodiky, kterou autorka promyšleně navrhla integrací několika zavedených přístupů. Její hlavní komponentou jsou testy shody relativních četností prvních a posledních číslic s pravděpodobnostmi podle Benfordova zákona. Autorce se podařilo dospět k podobným závěrům jako významným mezinárodním organizacím, jako je OBSE nebo Evropská unie, a to aplikací této navržené metodiky na prezidentské volby s různými politickými kontexty: Česká republika (2023), Spojené státy (2024) a Bělorusko (2020). Zatímco české a americké volby nevykazovaly žádné významné odchylky od očekávaných, běloruské záznamy odhalily četné nesrovnalosti, zejména v četnostech prvních a posledních číslic.

Klíčová slova

Benfordův zákon, první číslice, volební podvody, volební integrita

Contents

List of Figures

List of abbreviations

BL	Benford's Law
CZSO	Czech Statistical Office
EU	The European Union
ODIHR	OSCE Office for Democratic Institutions and Human Rights
OMV	Order of Magnitude of Variability
OSCE	Organisation for Security and Co-operation in Europe
USA	The United States of America

Mathematical symbols

α	the level of significance, the probability of the type II. error
β	the probability of the type II. error
D	a set of digits in a given position
d	a digit in a number
d_k	a digit in a number in the k position
d_L	a digit in a number in the last position
e	Euler's constant
G	a test statistic for the χ^2 goodness of fit test
H_0	a null hypothesis
H_1	an alternative hypothesis
j	a number of elements (digits) in the set D
k	a position in a number
λ	the parameter rate of the exponential distribution
m	a maximum value of the variable X
n	a number of observations
θ	a parameter
θ_0	a parameter coming from sample data
p	p-value
p_d	an observed relative frequency of a digit in a given position
π_d	a theoretical probability of a digit in a given position
T	a test statistic
t	a quantile of a theoretical distribution
V_α	an acceptance region
W_α	a rejection region
X	random variable from the exponential distribution
x	a value of a random variable X

Introduction

In an age where data-driven decision-making is increasingly central to public discourse, the integrity of numerical data, particularly in democratic processes such as elections, has never been more critical. One statistical phenomenon that has gained attention in this context is Benford's Law, a counterintuitive yet powerful tool for detecting digital anomalies in datasets. Originally observed in naturally occurring numbers, Benford's Law has found its applications in fields ranging from accounting to election forensics.

This thesis explores the application of Benford's Law to presidential election data to assess its effectiveness in identifying potential irregularities indicating record manipulation, one of many types of election fraud. the existence of election fraud poses a serious threat to electoral integrity, which is a cornerstone of any functioning democracy. Maintaining this integrity is essential to ensure that citizens can freely and fairly choose their representatives. To safeguard the democratic process, it is crucial to actively monitor and test for signs of electoral manipulation or fraud. Public trust in elections is not merely a procedural concern, it is fundamental to political stability and legitimacy. Without widespread confidence in the fairness of elections, democratic institutions risk erosion, and society as a whole becomes vulnerable to authoritarian tendencies.

The motivation for this research stems not only from growing public concern over electoral transparency but also due to the author's personal interest in both mathematics and political processes. Previous active involvement in the electoral process—first as a member of a Czech local election committee in 2020, and a year later as the head of the local electoral committee—provided valuable insight into how elections are conducted and where vulnerabilities may arise. These experiences, combined with a strong belief in the importance of electoral transparency, inspired the author to investigate whether mathematical tools like Benford's Law can contribute to the protection of election integrity.

By combining theoretical insight with empirical analysis, this thesis aims to:

1. **provide an introduction to Benford's Law for election fraud detection**,
2. **introduce a reproducible methodology for the detection of digital fraud in vote counts applying Benford's Law**, and
3. **showcase the use of the introduced methodology and compare the results with the findings of renowned organisations.**

To fulfil the goals, the thesis opens with an overview of the historical development of Benford's Law, its twice-in-history discovery and initial applications. It also explains how the law can be used to analyse election results and provides an overview of the electoral process. By the end of the first chapter, the reader is introduced to the terms election, election integrity and election fraud. Followed by the second chapter, which contains a theoretical overview of Benford's Law, including its mathematical formulation, key properties, and conditions for applicability. This is followed by an introduction to the statistical framework for hypothesis testing, a description of the data sources, and an outline of a practical workflow for applying the law to real-world data. The third chapter presents an analysis of three presidential elections: Czechia (2023), the United States (2024), and Belarus (2020), to evaluate the law's utility in diverse political and electoral contexts.

1. Benford's Law and Election Fraud

Benford's Law describes a particular behaviour of digits in a number. The lowest digits (such as 1 or 2) are most likely to occur in the first and second positions within a number. It has been demonstrated that this behaviour is common for numbers generated by various methods, including random linear combinations, aggregation of different datasets or random selection from such datasets, and processes arising from multiplication (such as geometric series or exponential growth). Such numbers can be found in all fields of science, including census data, stock prices, or the time intervals between earthquakes. (Cerqueti & Lupi, 2022; Hronová et al., 2023; Kossovsky, 2014)

While phenomena like this are beneficial for validation and fraud detection in many areas, it also helps to understand the discrepancy between the perceived and actual randomness in numbers. Many individuals believe that numbers are distributed evenly across numerous datasets or that repetition is minimal. An example of this: students were required to write out random numbers as part of an experiment, without prior knowledge of this law, they struggled to generate truly random numbers manually. (Beber & Scacco, 2012; Kossovsky, 2014)

This is also why various manual manipulations of numbers can be identified with high accuracy by BL. In an election context, the law is suited for detecting manual data manipulation. For us to analyse election data, it is necessary to understand where and how the manipulation can happen. First, its historical development will be explained, and then its use in the election context will be explained in this chapter.

1.1 History of Benford's Law

Benford's Law has been rediscovered multiple times throughout history. In fact, Frank Benford was not the first to identify this principle, yet it still holds his name. This section will describe who was the first to write about it, explain its historical development and comment on some of its initial applications. While the historical development of Benford's Law has no direct bearing on its practical application today, it is a fascinating demonstration of how simple numerical formulas can lead to statistically significant phenomena that occur across a variety of domains. This historical perspective helps to understand why and how the law became part of modern forensic statistics.

Simon Newcomb (1835-1909)

This notable Canadian-American astronomer and mathematician lived in the second half of the 19th century. And although he was recognised for his work in astronomy and physics, few people have noticed his article *'Note on the Fequency of Use of the Different Digits in Natural Numbers'*, where he had described the regularity in the occurrence of the first digits of a number, later known as Benford's Law. He demonstrates this phenomenon by calculating the relative frequencies of occurrence of the first and second digits as noted in equations 2.1 and 2.2. The behaviour of the remaining digits was only described by words, not by equations. (Kossovsky, 2014; Newcomb, 1881)

Frank Benford (1883-1948)

Frank Benford, after whom the law was named, lived at the turn of the 19th and 20th centuries. He was an American mathematician, physicist and electrical engineer, and held up to 20 patents in the field of optical devices. He was unaware of Newcomb's earlier paper on the frequencies of natural numbers when he published his much longer paper *'The Law of Anomalous Numbers'*. In contrast to Newcomb, Benford sought to further test the validity of the rule. Thus, he compared 20 large datasets of different data types and systematically recorded the results of these tests. However, his approach had flaws. For instance, it lacked an accurate mathematical description of the phenomenon, and some of his assumptions regarding the origin of the numbers were also unsubstantiated. His work was preceded by the famous story of the logarithmic tables - he saw that the tables had worn pages only at the beginning, but were as good as new at the end, as Newcomb had observed earlier, which made them both think about this phenomenon. Probably, those who used them most often needed to know the value of the lower numbers in their calculations more often than the higher values. (Benford, 1938; Hronová et al., 2023; Kossovsky, 2014)

The First Uses of Benford's Law in Practice

In 1972, Hal Varian, then a master's student at the University of California at Berkeley, came up with the idea of using Benford's Law to detect faulty data in economics. This is the first documented use of BL. He discovered a bug in a real estate agency's simulation program. So he started thinking about detecting faulty data - how to differentiate data that is *genuine* from flawed data in general. Previously, he had learned about BL. After applying the law successfully, he described it as *"almost numerological"*, since it was not as strongly mathematically justified at the time as it is today. He drew on practical results that suggested that the application was valid. (Kossovsky, 2014)

The first person to use the BL in a scientific paper was Charles Carslaw of the University of Canterbury in New Zealand in 1988. It was applied to some data from finance and accounting. He looked at how the earnings figures for New Zealand companies came close to the theoretical proportions of BL. It showed how, contrary to the BL for the last digits, earnings and income are often rounded to multiples of powers of 10, as well as the losses and expenses. (Kossovsky, 2014)

The 1990s then saw the addition of many more publications, notably by Mark Nigrini, C. W. Christian, S. Gupta and many others, who introduced the first forensic methods of how to use BL to detect manipulation or fraud. It is also worth mentioning Theodore Preston Hill (1995), who, relying on Benford's work, proved the theory of mixed distributions and added the necessary mathematical proofs.

Since then, this law is nowadays used as a standard test for many tax departments worldwide. Usually in the form of a routine test, where the first digits are compared to see how they are distributed across the entire dataset. Often, it is discovered that the digits are distributed evenly if the data is artificially created or has been poorly manipulated. It is also used in accounting, auditing and other financially oriented industries as well as in other fields with a risk of data fraud. (Kossovsky, 2014)

1.2 Use of Benford's Law in election context

While the use of Benford's Law in accounting and finance data in general is a prevalent practice, its use for validation of the election data is less evident. In contrast to financial records, electoral data may not immediately appear to align with the assumptions underlying Benford's Law. Nevertheless, as Kossovsky (2014) argues, any election data is a subset of a given population, which typically conforms to Benford's distribution. Moreover, the principle of scale invariance, discussed in detail in Section 2.1.2, supports the validity of applying Benford's Law to electoral datasets. This offers the potential to utilise the law as a diagnostic instrument for identifying fraud in reported election results.

Additionally, it is not advised to use Benford's Law as a stand-alone tool; rather, it should be complemented by other relevant tests. Lebeda et al. (2021) had analysed Czech election data from a quantitative and a qualitative perspective. The authors conducted an analysis of election turnout variance using regression models at an election office level and by interviewing members of election committees. The analysis of election fraud is a complex process, and the present paper focuses exclusively on one aspect of this complex phenomenon: namely, the detection of abnormalities in election counts.

In the field of digit-based diagnostics, academic literature has advocated an increased emphasis on the last digits for the analysis of manual counts, such as electoral results, as opposed to the first digits. Authors Beber and Scacco (2012) advise testing these biases in number generation according to psychology: the last digits should occur with equal frequency, and there should be some repetition among the numbers and digits. Note that these biases apply when fraud is done manually, and the fraud comes from a number generation done by a human. The BL is a suitable tool for detecting such manual record manipulation.

Deckert et al. (2011) suggests focusing on the last and next-to-last digits, since the fraud is expected to occur by rounding off the counts to tens and fives. However, he does not recommend using BL and the second-digit test for detecting election fraud, as Beber and Scacco (2012) described. Furthermore, the arguments of the latter have been disputed by the former (Mebane, 2011). It is asserted that the BL and the second-digit test can be useful if applied correctly and with proper theoretical grounding (Deckert et al., 2011).

1.2.1 Elections and Electoral Integrity

An **election** is a formal process of decision-making in which the population in question selects a representative or a group of representatives to hold public office or decide on a political proposition through a vote. Usually, elections are held in one day and are clearly announced in advance. The objective is to ensure that all individuals are granted an equal opportunity to participate. Voters proceed to designated polling stations, where their identity is verified and they are permitted to exercise their democratic right. The performance of the vote is subject to variation depending on the geographical location and the specific political context of the country in question.(Eulau et al., 2025)

Electoral integrity can be defined as a set of international and global standards that apply to the entire electoral process, including the periods before, during, and following an election, such as the counting and evaluation of election results. It is important to note that during this process, there is a possibility of misconduct and manipulation, which can be intentional or unintentional in nature. All of which add to the integrity of the election itself. (Eulau et al., 2025; Lebeda et al., 2021)

Electoral integrity can be maintained by a range of safeguards and procedural standards. These include maintaining a permanent, regularly updated voter list and ensuring that the registration process is accessible and straightforward if needed. Polling stations are typically managed by impartial government officials or clerks under official supervision. Political party representatives are permitted to observe the process, enabling them to monitor for irregularities and deter misconduct. Voting is conducted in private booths to ensure confidentiality, and ballots are counted—often with recounts—by designated tellers under the watchful eyes of party observers. Finally, the transmission of results from local polling stations to central election authorities is carefully secured and verified. Measures are also taken to preserve order at polling sites, such as deploying police. (Eulau et al., 2025)

Transparent elections are considered to be the cornerstone of democracy. The credibility of legitimate and transparent elections is also linked to society's confidence in democracy and democratic principles, which populist and authoritarian movements and parties like to exploit. It is therefore imperative that efforts are not directed towards discrediting previous electoral processes; rather, the implementation of analytical tools could serve to dispel any such doubts in a more efficient and expeditious manner. (Eulau et al., 2025; Lebeda et al., 2021)

1.2.2 Election Fraud

Election fraud, manipulation or voter fraud and voter rigging are all names for electoral abuse. These practices, which have been deemed dishonest, are intended to influence the results of elections, often to subvert democratic processes. It is evident that these phenomena are in direct opposition to the concept of electoral integrity. The legitimacy of electoral processes can be impacted by these factors, which can result in a decline in voter turnout or an escalation in distrust and a diminution in support for democratic political processes. Some of these practices can be legal, but morally unacceptable. It is important to note that while some of these practices may be legal, they are ethically questionable and, as such, should be avoided. It is imperative for public institutions to be proactive in detecting and safeguarding a democratic society, with a particular focus on preserving public trust in free and fair democratic elections and their outcomes. (Eulau et al., 2025; Lebeda et al., 2021; Lehoucq, 2003; Levitt, 2007)

The definition of election fraud is a bit complicated. Levitt (2007) noted that using the term *voter fraud* has the potential to engender the perception that the responsibility for such actions lies with the individual voter. Lehoucq (2003) concentrated on systematic influences that may change the voter's decision, such as voter intimidation or bribery. However, these are not the focus of this thesis, as they can not be detected by digital methods. That is why in this thesis, the authors have used the term election fraud exclusively. In so doing, the focus is shifted towards the election system in its entirety, from the process of collecting and counting votes to their subsequent entry into a database and analysis.

Election fraud can be defined as a purposeful manipulation of an election process, typically conducted under the cloak of secrecy, by influencing either the electorate or the results of the election itself. In this thesis, the focus will be on fraud that can be diagnosed digitally, namely, record manipulation. (Lebeda et al., 2021)

Record manipulation is defined as a fraudulent practice of altering the vote count. This is typically executed in the last phase after the election, following the counting process and the collection of results. The vote count for a particular candidate is rewritten, with the number being adjusted to achieve the desired proportion or result; alternatively, the generation of entirely new election results based on the desired outcome. In the absence of prior knowledge of BL, the process can be detected by simple tests, as shown in this thesis later on. (Kossovsky, 2014; The Heritage Foundation, 2024)

2. Theoretical Background and Methods

Having explored the historical development of Benford's Law from its first applications in economics and accounting to its more recent role in assessing electoral integrity, we now turn to the methodological framework that underpins its application. This includes a description of Benford's Law, its mathematical formulation, and the probability distribution of leading digits and its properties, like the scale invariance principle. Moreover, to evaluate whether a dataset adheres to Benford's distribution, we must employ statistical techniques, mainly hypothesis testing. After selecting a suitable test and proposing an analysis workflow, we should be ready for the analysis in the next chapter.

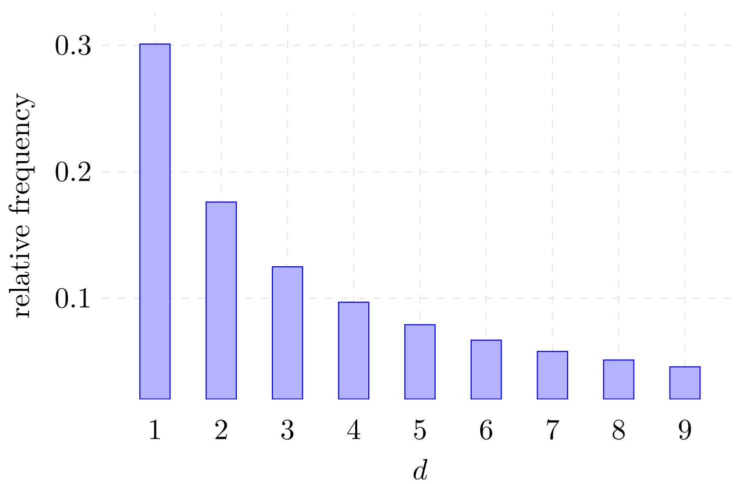
2.1 Benford's Law Formulation

Benford's Law states that lower digits like 1 and 2 appear more frequently at the beginning of numbers in many naturally occurring datasets. According to Benford's first digit law, the **relative frequencies of the leading digits** are approaching

$$\mathrm{P}(X = d) = \log_{10}(1 + 1/d), \tag{2.1}$$

where $d \in \{1, \ldots, 9\}$ for first digit law and $d \in \{10, \ldots, 99\}$ for the first-two digits law or even $d \in \{100, \ldots, 999\}$ for the first-three digits law. For other values (negative d), the probability is 0. When plotted, it resembles a logarithmic distribution, as seen in Figure 2.1. (Cerqueti & Lupi, 2022; Hronová et al., 2023; Newcomb, 1881)

Figure 2.1: Theoretical frequencies of the first leading digits



Source and processing: author

The first leading digit is the first non-zero number at the highest order. In the case of the number 124, the first digit would be 1, whereas for the number 0.039, the first digit would be 3. According to Benford's Law of the first digit, the theoretical frequencies correspond to the equation 2.1, indicating that the digit 1 occurs in the first position with a probability of 0.3, the digit 2 with 0.18, and so forth, with the digit 9 occurring with a probability of only 0.046. For probabilities of all first digits from 1 to 9, see the Table 2.1.

Table 2.1: The probabilities of the first leading digits according to the BL

d	1	2	3	4	5	6	7	8	9
$\mathrm{P}(X = d)$	0.301	0.176	0.125	0.097	0.079	0.067	0.058	0.051	0.046

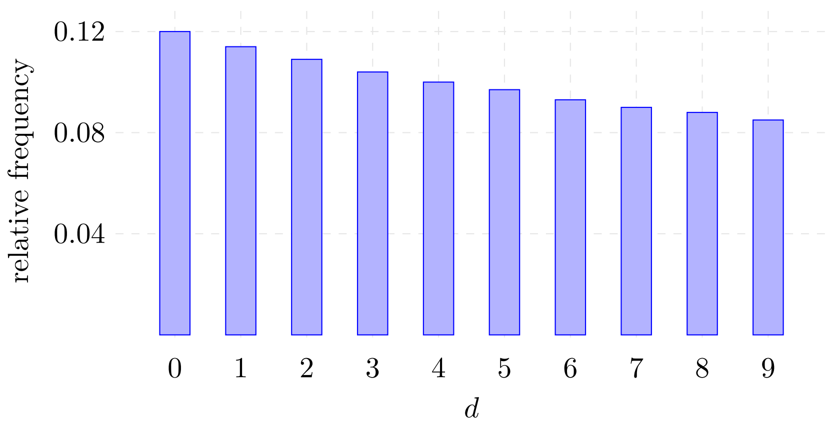
Source and processing: author

As seen in the Figure 2.1, the distribution is heavily skewed in favour of the lowest digits. As we move to the second and higher digits, this skew will flatten out significantly until there is no difference between the digits. (Kossovsky, 2014) the **relative frequencies of the second leading digits** according to the Benford's Law approaches the probability

$$P(d) = \sum_{k=1}^{9} \log_{10}\left(1 + \frac{1}{10k + d}\right), \quad \text{for } d = 0{,}1, \ldots, 9 \tag{2.2}$$

as mentioned by Hronová et al. (2023), while assuming the independent occurrence of the second leading digits. The second leading digit is the second number from 0 to 9 at the highest order of a number. The distribution of the second leading digits is less skewed than that of the first leading digits, while maintaining the trend of favouring the lower digits over the higher digits. The probabilities according to Benford's Law of the second leading digit can be seen in Table 2.2.

Figure 2.2: Theoretical frequencies of the second leading digits



Source and processing: author

Table 2.2: The probabilities of the second leading digits according to the BL

d	0	1	2	3	4	5	6	7	8	9
P(d)	0.110	0.114	0.109	0.104	0.101	0.097	0.093	0.090	0.088	0.085

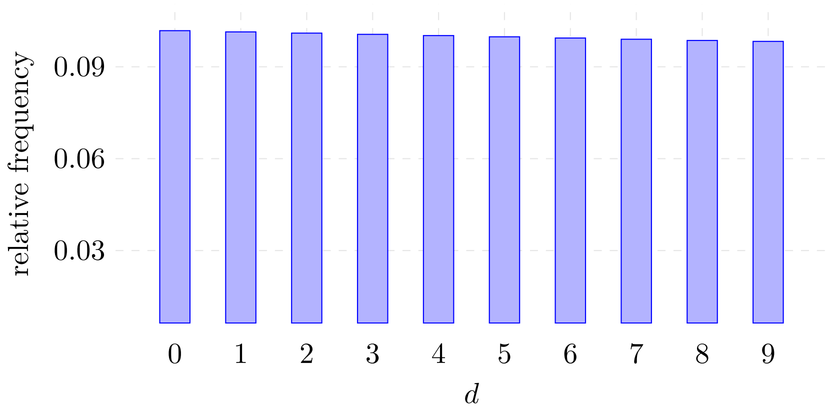
Source and processing: author

Coming to the third leading digit, the relative frequencies begin to be close to equal as seen in Figure 2.3. This tendency is explained by the following equation of the **generalised Benford's Law**:

$$P(d_k) = \sum_{d_1=1}^{9} \sum_{d_2=0}^{9} \cdots \sum_{d_{k-1}=0}^{9} \log_{10}\left(1 + \frac{1}{\sum\limits_{i=1}^{k} d_i \cdot 10^{k-i}}\right), \quad \text{for } d_k = 0{,}1,\ldots{,}9 \tag{2.3}$$

describing all leading digit relative frequencies given their leading position, again with the assumption of independence of digit occurrences. (Hronová et al., 2023) According to which the probability of lower digits is higher for the leading digits, rather than for the later digits in a number.

Figure 2.3: Theoretical frequencies of the third leading digits



Source and processing: author

To get the probabilities of the third leading digits, we need to sum all the probabilities according to Formula 2.1 of all the possible numbers with the given digit in the third position. Illustrated by calculating the probability of number 6 being in the 3rd leading position (denoted as6_3) by the following

$$P(6_3) = \log_{10}(1 + 1/106) + \log_{10}(1 + 1/116) + \cdots + \log_{10}(1 + 1/996).$$

Summation of the probabilities causes the differences to shrink. and so does the distribution converge to a uniform distribution. In conclusion, for the digits in the last positions, the distribution is expected to be very close to, if not uniform. This convergence has been shown by Hill (1995). As can be seen in Table 2.3, when compared to the first and second leading digit probabilities in Tables 2.1 and 2.2, the probabilities are much closer to such probabilities as if they were uniform.

Table 2.3: The probabilities of the third leading digits according to the BL

d	0	1	2	3	4	5	6	7	8	9
P(d)	0.102	0.101	0.101	0.101	0.100	0.100	0.010	0.099	0.099	0.098

Source and processing: author

2.1.1 Other relevant distributions

Although Benford's Law exhibits a distinctive logarithmic distribution of digits, it is useful to compare it with other statistical distributions to better understand its uniqueness and limitations. In the following section, we will briefly introduce the uniform and exponential distributions. The uniform distribution represents a scenario where all digits occur with equal probability. (Hill, 1995) Additionally, the exponential distribution is briefly introduced—not because it shares theoretical foundations with Benford's Law, but because its shape can, under certain parametrisations, resemble the digit distribution of Benford's Law. Including it here helps underscore the importance of not conflating visual resemblance with conceptual relevance.

Distribution, where every value of a random discrete variable has the same probability, is named the **uniform distribution**. It has many uses, for example, it can select a value from a set when all values have the same probability. In the context of this thesis, the random variable d_L is a digit in the last position of a number. The probability is

$$\begin{aligned} P(d_L) &= \frac{1}{m}, \qquad && d_L = 0,\ 1,\ \ldots,\ 9 \\ &= 0, && \text{else,} \end{aligned} \tag{2.4}$$

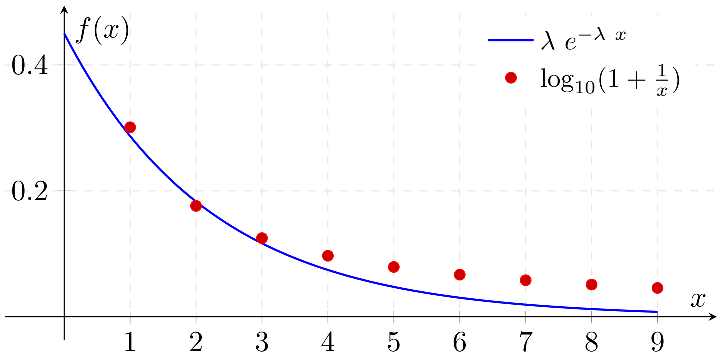
while m is the number of all possible values of d_L. (Marek, 2024) in the context of this thesis, $m = 10$ for the case of the last digits. If digits in a long enough number were distributed equally, according to the uniform distribution, all digits would have the probability of 0.1. (Hill, 1995; Hronová et al., 2023; Kossovsky, 2014)

The **exponential distribution** is similar in shape to the BL first leading digit distribution, as seen in Figure 2.1. The exponential probability distribution function for a random variable X is defined in Equation 2.5, where λ is the rate parameter. (Marek, 2024)

$$f(x;\lambda) = \lambda\, e^{-\lambda\, x} \quad \text{for } x \geq 0 \tag{2.5}$$

These two distributions may look similar when selecting the appropriate value of the parameter λ, as can be seen in Figure 2.4, where the λ parameter was set deliberately to fit the exponential distribution to the distribution of the first digits according to Benford's Law. Nevertheless, their uses and meanings differ significantly.
The exponential distribution describes the distance between independent events in time, whereas Benford's digit distribution assigns the probability to a given digit in a given position. (Marek, 2024)

Figure 2.4: Benford's first digit law and the exponential distribution
with parameter $\lambda = 0.45$



Source and processing: author

2.1.2 The Scale Invariance Principle of Benford's Law

Exploring the mathematical properties of Benford's Law, one of the most essential is the scale invariance principle. This property explains why BL holds across datasets measured in different units or scales, reinforcing its utility in diverse analytical contexts. Given a dataset that behaves according to BL, changing units (which arithmetically means multiplying the whole dataset by the same constant) will still behave according to BL. Any arithmetic operation applied to the whole dataset does not change the underlying BL distribution. (Hronová et al., 2023; Kossovsky, 2014)

2.1.3 Digital Development Pattern

Beyond scale invariance, another intriguing characteristic of Benford's Law is its manifestation across varying magnitudes of data. This phenomenon, known as the digital development pattern, refers to the way distributions of digits evolve across different ranges of data. This pattern is observed in all random data sets, whether they follow Benford's Law or not. The pattern shows how the distribution of digits changes from lower to higher values of the data range.

This pattern can only be seen when the data range is partitioned along integral powers of ten, such as (0.1, 1), (1, 10), (10, 100), etc. The pattern shows increasing skewness as you move from the left to the right of the data range. For smaller values, digit distributions are more equal. In the centre, the distribution becomes more logarithmic. For larger values, the distribution becomes highly skewed in favour of low digits. (Kossovsky, 2014)

2.1.4 Applicability of Benford's Law

While the mathematical formulation and theoretical properties of Benford's Law provide a strong foundation for its use, it is equally important to understand the practical conditions under which the law can be reliably applied. The law does not impose many conditions on the data that will be analysed, yet some must be met for reliable results. This section will discuss the following: sufficient variability, appropriate sample size, and the exclusion of certain types of values.

Order of Magnitude of Variability

One of the distinct conditions for data to be analysed is the variance of the data. And that is done by the condition for the order of magnitude of data variability. Kossovsky notes that for data to behave according to BL, its value range should be wider, rather than narrow. He describes a measure, Order of Magnitude of Variability (OMV), as

$$OMV = \left(\log_{10}(90\text{th percentile}) - \log_{10}(10\text{th percentile})\right) > 3 \tag{2.6}$$

and recommends it should be larger than 3. In practice, this means that our data should spread from 1 to 10,000 or more for consistent results. If we were to use data with lower OMV than recommended, we would get results that look fraudulent, while being genuine, data like this has higher probabilities of type I error when testing for compliance with BL. However, election data in general comply with the BL distribution. (Kossovsky, 2014)

Varying results from the expected distribution

It is generally advisable to focus on *deviations* from the theoretical distribution in a given BL variation, specifically on the significantly higher values for the frequency of occurrence of a given digit. In graphical representation, these can be represented as spikes. There is minimal interest in the reduced frequencies, as any suspicious activity is likely to manifest itself through increased frequencies - when counting frequencies, an increase in one digit is much more detectable, with all other frequencies dropping slightly. (Kossovsky, 2014)

Deviance is to be expected, but where should the line be drawn? Kossovsky suggests distinguishing between two terms: compliance and comparison. When researching the data's **compliance** with BL, the data is expected to follow the BL distribution very closely, the focus is on detecting manipulation, and if there is, whether it is random or structural. For the use in this thesis, we assume *the data follows the distribution* (as explained in section 1.2) and therefore observe data compliance.

Next, the deviations should not show signs of any systematic error or *pattern*. Some pattern may emerge by rounding numbers, for example. The practice of rounding data is prevalent in the context of accounting, where values ending in 00 or 50 are frequently observed. this rounding process, which is often employed to streamline computations, can give rise to manipulation of higher orders while being harmless in context. Empirical evidence supports the prevalence of these values in practice. However, such manipulations are unacceptable in, for example, electoral statistics, where votes simply cannot be rounded. (Deckert et al., 2011; Kossovsky, 2014)

Alternatively, the validity of BL can be influenced by changed external situations in regards to the data itself. For instance, how does sales data follow BL distribution when affected by the pandemic in 2020? this has been answered by Hronová et al. in 2023. Changes like this should be taken into account, and analysis should be adapted accordingly.

Correct usage

The compliance of our data with Benford's Law is to be expected when only minimal conditions are met. When using Benford's Law to detect intentional fraud or data manipulation, the *complete dataset* is used, rather than a subset. The *dataset should be large enough* - for datasets with less than 100 observations, BL analysis is not recommended. Only *non-negative data* should be used for analysis. For use in auditing, it is also recommended to eliminate small numbers, values less than, for example, 50, but that the portion removed be less than 10%. In addition, values that represent totals or summations should not be included if they come from the same sample.
(Cerqueti & Lupi, 2022; Kossovsky, 2014)

A dataset fit BL compliance testing :

- has more than 100 observations
- has all values ≥ 0
- has no summations or totals included in the dataset
- is a complete dataset rather than a subset
- is of population nature (eg. census or election) or OMV > 3.

2.2 Hypothesis testing

Having established the theoretical foundations and practical considerations of Benford's Law, the next step is to assess whether a given dataset conforms to its expected BL distribution. This is where hypothesis testing becomes essential.

Hypothesis testing is a fundamental framework used to test a theory about a population based on a sample by comparing two hypotheses in statistics at the given **level of significance** α. The tested hypothesis is the *null hypothesis* H_0. An alternative to this, usually a negation, always a disjunction, is the *alternative hypothesis* H_1.

$$H_0 : \theta = \theta_0 \qquad H_1 : \theta \neq \theta_0 \tag{2.7}$$

In the case of the hypotheses in 2.7, the parameter θ is tested whether its value is θ_0, which comes from sample data. Based on the alternative hypothesis, this test can be a one-sided or two-sided test problem. The alternative in this equation is the **two-sided test problem**. If changed to $\theta < \theta_0$ or $\theta > \theta_0$, this would be the **one-sided** test problem. (Heumann et al., 2022; Malá, 2024)

To test this, a sample from the population is taken and analysed. Given the nature of the test, the **test statistic** T is selected. Its value can be in the **rejection region** W_α, and conflict with the null hypothesis, or the value is in the acceptance region V_α, and there is no conflict with the null hypothesis. The **critical value** is the boundary between the two regions and for rejecting the null hypothesis coming from the distribution under the H_0 t. In the end, it is decided based on the critical value and test statistic whether the null hypothesis is rejected. Or the **p-value** is computed and compared to the level of significance. (Heumann et al., 2022; Malá, 2024; Marek, 2024)

Type I. and Type II. errors

When conducting hypothesis testing, it is crucial to understand the potential outcomes and their implications. There are two specific kinds of errors, and the probability of one influences the probability of the other. Their description will be covered in the following paragraphs.

Type I. error α is the given level of significance, and it is the probability of rejecting the null hypothesis if the null hypothesis is correct. the α is usually set at 0.05.

$$\alpha = P(W_\alpha | H_0) \tag{2.8}$$

Type II. error β is the probability of not rejecting the null hypothesis if the null hypothesis is not correct.

$$\beta = P(V_\alpha | H_1) \tag{2.9}$$

The probability $1 - \beta$ is the **power of the test** and denotes the probability of correct rejection of the null hypothesis. The two errors are not equivalent from the perspective of the significance test. the α is maintained, while β is minimised. Both errors can only be minimised by increasing the number of observations. (Heumann et al., 2022; Malá, 2024)

P-value

One way of deciding a statistical test, much more common in statistical software, is by the p-value. The p-value is defined as "*the probability of obtaining a result which is as* *far or further from H_0 than the observed value*". The decision is made by comparing the p-value to the level of significance α. We reject the null hypothesis when $p < \alpha$ and we reject the H_1 when $p \geq \alpha$. (Malá, 2024) the p-value is defined as

$$\text{two sided case: } P(|T| \leq t|H_0) \tag{2.10}$$

$$\text{one sided case: } P(T \geq t|H_0) \tag{2.11}$$

$$P(T \leq t|H_0) \tag{2.12}$$

as referenced by Heumann et al. (2022).

2.2.1 χ^2-goodness of fit test

To check the compliance of the data to the BL, we use the χ^2 goodness of fit test. This test is generally used to test whether the observed data comes from an expected distribution. In this test, the observed frequencies are compared to the expected frequencies. The null hypothesis assumes the empirical distribution follows the theoretical distribution, the alternative hypothesis is the opposite, as defined in:

$$\begin{aligned} H_0 &: \pi_d = p_d, \qquad d \in \ D \\ H_1 &: \pi_d \neq p_d, \end{aligned} \tag{2.13}$$

d comes from a set D, which contains all possible digits. For testing the compliance of the first leading digits in numbers according to BL, $D = \{1,2,\ldots,9\}$. j is the number of elements in the set D, in this case, $j = 9$. If we were testing the second and next in-order digits, the set D would have 10 elements.

The test statistic is given as such

$$G = n \sum_{d=1}^{9} \frac{(p_d - \pi_d)^2}{\pi_d} \qquad\qquad G \approx \chi^2(j-1) \tag{2.14}$$

where π_d is the theoretical frequency under BL distribution,
p_d is the empirical frequency from the data
n is the sample size, and
d is the corresponding digit

as recommended in Hronová et al. (2023) and Kossovsky (2014).

There is one assumption, and that is that the sample size is large enough. $n \cdot \pi_d > 5$ for all d (as put differently, there are enough observations for each digit). The rejection region under the null hypothesis is

$$W_\alpha \ = \left(\chi^2; G \geq \chi^2_{1-\alpha}(j-1)\right) = (\chi^2_{1-\alpha}(9-1), \infty) \tag{2.15}$$

for the distribution of the first leading digit. The test can be summed up as the larger the fraction 2.14, the poorer the fit between the empirical and theoretical frequencies.

2.3 Data sources

From the fundamental theoretical concepts and definitions to implementing gained knowledge into practice. In the first step, this process involves the selection of appropriate election data. In this thesis, we analyse the results of presidential elections from three different countries - **Czechia, the USA** and **Belarus**. The decision to include these countries is based on their unique context and relevance to the goals of this thesis, which will be expanded on in subsequent sections. Czechia is familiar to the author, the USA presents a different electoral system, and elections in Belarus are considered to be fraudulent (Council of the EU, 2020). All of these datasets present a challenge to the methodology used.

Presidential elections were chosen for this analysis due to their relative simplicity and transparency compared to other types of elections. In most countries, presidential elections are direct votes, where the candidate with the majority of votes wins, making the process straightforward to describe and analyse. Unlike parliamentary elections, which often involve complex mechanisms such as proportional representation, coalition-building, and multi-round seat allocations, presidential elections typically avoid these complications. (Eulau et al., 2025). This reduces the number of stages where manipulation could occur, allowing for a more direct interpretation. Additionally, the structure of presidential elections facilitates easier comparison across countries. This said, the election system is a little different in each country. In the subsequent sections, the differences between the countries are highlighted, and their socio-economic context is explained.

2.3.1 Czechia 2023

The first case is the 2023 presidential election in Czechia. The author is from the Czech Republic, making the electoral system and political context more familiar and easier to interpret within the scope of this work. in the Czech Republic, the presidential elections have two rounds, after which the president is elected for 5 years. The votes are counted in polling stations, and the counts are directly sent to the CZSO. The results are published from there; the system is quite centralised. (Lebeda et al., 2021; OBSE, 2023; Zákon č. 88/2024 Sb. o správě voleb, 2024)

This data is easily accessible through the CZSO website (2023), even for individual cities in Czechia, or by accessing the data through the catalogue of the Open Data from the CZSO using the `czso` R package (Bouchal, 2024) on the individual election districts level. The **vote counts per municipality in the second round** were selected. In total, there are 6,254 observations, each representing a municipality, by total vote counts and the vote share of a candidate. This dataset was expanded by the turnout tables for municipalities. These tables contained the turnout, number of eligible voters, the number of valid votes and the number of issued envelopes and the number of returned envelopes. The foreign votes and turnout were manually added from the regional datasets (both vote counts and the turnout tables). And after this, the aquired data was ready for analysis.

The Czech electoral practice has its weak points from the point of view of electoral integrity, as described and analysed by Lebeda et al. (2021); however, during the 2023 presidential election, there were no suspicions about election fraud as such or any significant malpractice regarding the gathering of the election results or record manipulation by our knowledge and media coverage. The OBSE election observation (2023) assessed the election campaign in the media and its finances. But the report for the previous election by ODHIR (2018) suggested there is no evidence of election fraud and that the election process is fair and competitive. There have been no significant changes in the election process since then.

After the election, there were more than 400 complaints about the results and the election process to the Czech Supreme Administrative Court (2023). This is a common part of the democratic election system, as anyone has the right to complain and ask for an investigation within two days after the election, according to the Czech law (Zákon č. 88/2024 Sb. o správě voleb, 2024). After the investigation, the election results were slightly changed (by adding about 300 votes for Petr Pavel (Nejvyšší správní soud ČR, 2023)), but the final results are not invalidated, and the election integrity is maintained. This edited version of election counts is analysed.

2.3.2 USA 2024

The second case is focused on the 2024 presidential election in the USA, which attracted significant global attention. Some allegations of electoral manipulation circulated in the public, impacting public trust and making it a compelling subject for digital forensic analysis using Benford's Law. However, the claims about election fraud were unfounded. (OBSE, 2025)

The United States do not have a central statistical service that would provide the nationwide data on a county level, which we need for this analysis. A county is a subdivision of a U.S. state. In the USA, according to the ODHIR (2025), the system is deeply decentralised. The official data from each state is available separately on the state's website. However, McGovern et al. (2025) was able to scrape data from the Fox News website, which provided us with the vote counts for each county as needed. In this dataset, there were only the vote counts of the two main candidates, Donald Trump and Kamala Harris. This data already came prepared for analysis, so there was no need for further manipulation. Due to the nature of the election process in the USA, there is no data on the voter turnout, and unfortunately, this dataset does not contain the shares of valid votes either.

Each USA state performs its election process according to the state law, and then the results for the entire country are summed up. Then, the votes do not go directly to the one candidate, but rather to the electoral college that votes for the candidate.

(USA.gov, 2025) this difference, however, does not affect the performance of the BL analysis due to the scale invariance principle (as described in section 2.1.2) and because, no matter the rules, the selection of the votes is still a fraction of the population, so we should be able to get reliable results.

2.3.3 Belarus 2020

In the third case, we examine the 2020 presidential election in Belarus. Although the author initially intended to analyse the most recent Belarusian election, the necessary data was not made publicly available, likely due to the authoritarian nature of the regime and efforts to suppress evidence of electoral fraud. Fortunately, data from the 2020 election is accessible and provides a valuable opportunity to explore potential irregularities in a highly contested political environment.

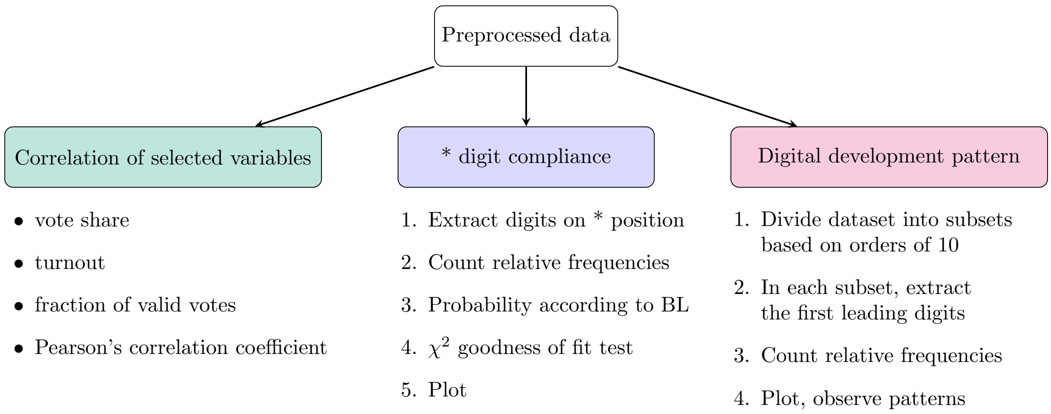
Honest People and Zubr are civic organisations of the Belarusian opposition. Their members had gathered and processed **the final protocols of 1,310 polling stations** out of 5,767 throughout Belarus. And these protocols were made available online on Kaggle (2020). Since the official data was not made public, a sample must suffice for our analysis. This is fine with the BL, since this sample is a fraction of the population, so the law should apply anyway, thanks to the scale invariance principle (section 2.1.2).

In this election, there were 5 candidates; however, due to their low vote shares, this analysis is focusing mainly on the two most popular candidates - Alexander Lukashenko, former leader and his opponent, Sviatlana Tsikhanouskaya, who took over after her husband was arrested. These elections were controversial. Demonstrations were happening before and long after the elections. The Belarusian society has awakened in the whole country. Before the elections, some of the candidates were intimidated. The democratic principles of an election were not fulfilled. The ODIHR were not invited to observe the election and inspect the election integrity. These elections are considered fraudulent; their results have not been accepted by many democratic states and have caused the EU to impose sanctions on the country. (Bedford, 2021; Council of the EU, 2020; OBSE, 2020a, 2020b)

2.4 Proposed steps of the analysis of the Benford's Law compliance

After selecting the data, the next step is to outline a structured approach for applying Benford's Law in practice. The following section presents a step-by-step procedure for an analysis applying Benford's Law. An organised workflow can be seen in Figure 2.5. Its elements, from data preparation to interpretation of acquired results, will be commented on below.

Figure 2.5: Proposed workflow for analysing election results and applying Benford's Law



Source: author

2.4.1 Data preprocessing

The election results for small territorial units are chosen - one statistical unit is a town in the Czech Republic, a county in the USA or a polling station in Belarus. Selection or adaptation of the size of the territorial unit is based on individual accounting for the size of the country, total sample size and availability of the data, so that the dataset would be suitable for the methodology of this thesis, namely the BL, as stated in section 2.1.4. Additionally, the rows of totals are removed if present in the data.

Next, the relevant variables are selected. Most important are the vote counts of the candidates and the totals. There are two candidates in the Czech elections, the USA and Belarus have two main candidates and a couple of other candidates with very few votes. The two candidates with the majority of votes are selected. Afterwards, the data types are changed into those that are suitable for the analysis; vote counts are typically a numerical discrete (integer) variable, and their validity is checked (control sums of the number of votes).

2.4.2 Correlation between selected variables

As Lebeda et al. (2021) mentioned, observing the relationships between the valid votes and the votes for each candidate is an essential part of election analysis, so this part shall not be forgotten. It is not in the primary focus of this thesis, however, it will add an additional point of view in this analysis. We are on the lookout for some patterns and a high Pearson's correlation coefficient between turnout and votes for a specific candidate, or any anomalies of other kinds. Ideally, the turnout, the percentage of valid votes, should both be uncorrelated with the vote share. Unusually high support in high-turnout areas can suggest the presence of non-digital election fraud.

The patterns can be observed through scatter plots, a graphical way to not only visualise correlation between two variables but also to be able to observe patterns and other non-linear relationships among the data. There are only two **clusters** expected: first, in the shape of a vertical line in a plot where votes and valid votes are on the axes. Showing polling stations that have all the votes valid, and second, horizontal, for the unsuccessful candidates, whose vote shares are close to zero. Other than that, there are no expected clusters.

2.4.3 Digit compliance on a given position

In this most important part of the analysis, we look at how well the relative frequencies follow the BL variation. Namely, we shall focus on these positions in numbers:

- first digit,
- first two digits,
- second digit,
- third digit, and
- last digit.

As Kossovsky (2014) recommends, observing the BL in more variations, not just Benford's first digit law, provides a more comprehensive insight into the digit distribution compliance.

In the first digit distribution, the BL is applied, as formulated in Equation 2.1. The same information, but in better detail, is seen in the distribution of the first two digits. The second digit distribution, as noted by Beber and Scacco and Mebane, is suitable for detecting digital fraud. Similarly, with the third digit distribution, for which the distribution starts to resemble the uniform distribution, as can be seen in Figure 2.3. And lastly, we shall take a look at the relative frequencies of the last digits with expected uniform distribution as in 2.4.

To do all this, first, the relevant digits in the given position need to be extracted. Then their relative frequency and the probability according to the BL are counted, as seen in an example table 2.4. This table is used for the following testing and plotting.

Table 2.4: The frequencies of the first leading digits and probabilities according to the BL the Czech presidential elections 2023

digit	Freq	Total	RelFreq	BL	diff
1	1866	6320	0.2953	0.3010	-0.0058
2	1090	6320	0.1725	0.1761	-0.0036
3	811	6320	0.1283	0.1249	0.0034
4	621	6320	0.0983	0.0969	0.0013
5	492	6320	0.0778	0.0792	-0.0013
6	434	6320	0.0687	0.0669	0.0017
7	368	6320	0.0582	0.0580	0.0002
8	302	6320	0.0478	0.0512	-0.0034
9	336	6320	0.0532	0.0458	0.0074

Source: data: Czech Statistical Office (2023), processing: author

All those distributions for given positions shall be tested by the χ^2 goodness of fit test, as defined in 2.14, and then plotted with the corresponding p-value of the test. In some graphs, the p-value is labelled as *unreliable*, which signifies that the assumption of the test was not met (the absolute frequency of any digit was smaller than 5).

There are also two points of view, the first on the totals of votes and the second on the dual plots with the counts for the individual candidates. This is important to see if there was an interest in manipulating one candidate more than the other.

The distribution of the last two digits is not examined due to inconclusive results given by the nature of the data, the range is from 0 to 10000 for most of our data, depending on the country and the examined regional level, and for some numbers the last two digits can be the whole number, so the relative frequencies do not match.

2.4.4 Digital development pattern

Tn this last part, the digital development pattern is visually inspected. The expected behaviour of such is described in a more detailed manner in section 2.1.3. Generally, we are searching for some differences from expected behaviour while also observing what part of the dataset is represented in a given range. This provides additional information on the genuineness of the data.

To be able to observe the pattern, additional conditions are imposed on the data. The range of the values should be larger. It may be a bit problematic, because election districts are constructed in a way to have the same number of possible voters. This specific situation occurred in the case of the Belarusian data, where a simple aggregation to cities from polling stations was necessary to enable the observation of the pattern.

First, the dataset must be divided into parts by the values, the border points of the intervals marking the integral powers of ten. Then count the relative frequencies of the first digits in each part, as described for digit compliance on a given position in a section 2.4.3. The number of observations and the total share of each part must be included in the output table. An example of relative frequencies of the digits in different parts of the dataset can be seen in Table 2.5. This table values are then plotted and visually evaluated.

Table 2.5: Digital development pattern table of the Czech presidential elections 2023

values from	1	10	100	1,000	10,000
values to	10	100	1,000	10,000	100,000
1	0.024	0.057	0.466	0.578	0.667
2	0.000	0.113	0.215	0.169	0.146
3	0.048	0.139	0.116	0.094	0.104
4	0.024	0.129	0.072	0.049	0.000
5	0.071	0.129	0.043	0.036	0.021
6	0.095	0.125	0.034	0.036	0.042
7	0.238	0.112	0.024	0.015	0.021
8	0.143	0.104	0.017	0.011	0.000
9	0.357	0.091	0.012	0.011	0.000
# of data points	42	5,106	6,595	716	48
% of data	0.336	40.825	52.730	5.725	0.384

Source: data: Czech Statistical Office (2023), processing: author, inspired by Kossovsky (2014)

3. Application on Election Data

Having established the theoretical background of Benford's Law and election fraud and the statistical framework of our analysis, the final step is to apply this knowledge to real-world election data. This chapter demonstrates how the concepts discussed throughout the thesis can be implemented in practice, using selected datasets, as described in section 2.3, to evaluate conformity with Benford's distribution.

Three cases were selected to illustrate the application of Benford's Law under different electoral and political conditions: the Czech presidential election in 2023, the USA presidential elections in 2024, and the Belarusian presidential election in 2020. Through this application, we aim to showcase Benford's Law as a tool for detecting anomalies and drawing meaningful conclusions from empirical data. By comparing the statistical behaviour of digits across these three elections, we aim to highlight both normal and anomalous patterns and to contextualise them within the political and procedural settings in which the elections took place.

The analysis was performed in the R software (R Core Team, 2023), using the following libraries: `read` (Wickham, Hester, & Bryan, 2024), `stringr` (Wickham, 2023), `dplyr` (Wickham et al., 2023), `ggplot2` (Wickham, 2016), `tidyr` (Wickham, Vaughan, & Girlich, 2024), `patchwork` (Pedersen, 2024), `czso` (Bouchal, 2024). The source code can be accessed on GitHub (2025).

3.1 Presidential elections in Czechia in 2023

As a first empirical application of Benford's Law, we examine the Czech Republic's 2023 presidential election. This two-round, direct popular vote system presents an ideal test case: it uses centralised vote reporting, lacks an electoral college, and our data is aggregated at the municipality level.

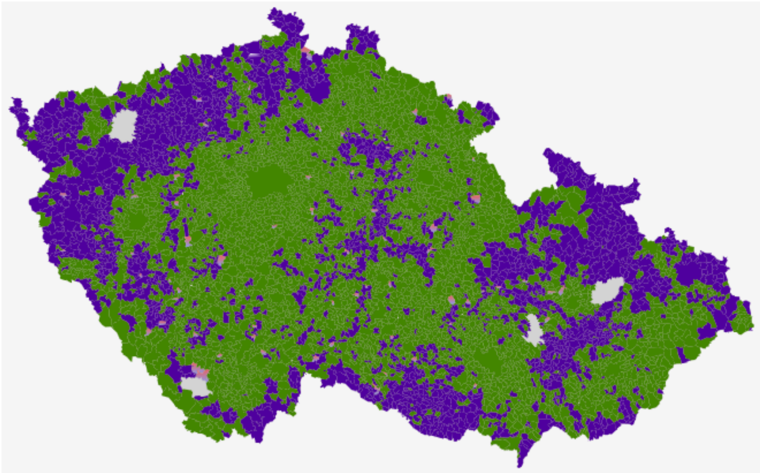
The first election round took place from 13th to 14th January. The second election round on 27th and 28th January 2023 saw Petr Pavel defeat Andrej Babiš by a margin of almost a million votes. The final vote distribution on the municipality level can be seen on the map in Figure 3.1, and the final results in Table 3.1.

Table 3.1: Election results in Czechia 2023

candidate	votes share
Petr Pavel	58.32%
Andrej Babiš	41.67%

Source: Czech Statistical Office (2023)

Figure 3.1: The results of the presidential elections on the city level in Czechia in 2023



Source: Czech Statistical Office (2023)

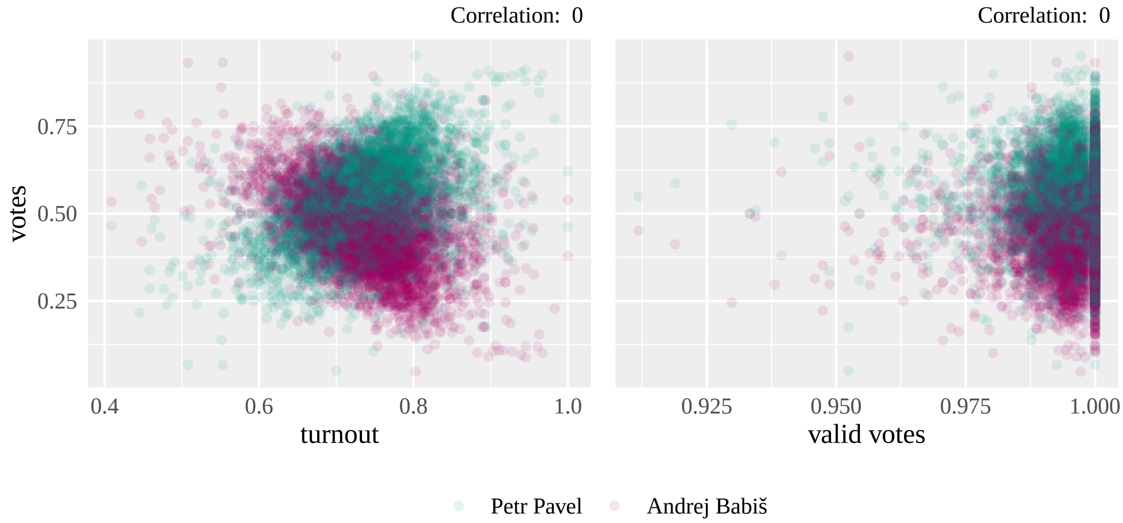
In the map, Petr Pavel is signified by green, and Andrej Babiš is signified by purple. The pink territories are where the vote share was equal, grey is a territory without a ward. Turnout was 70.25% of eligible voters. 99.51% the votes were valid. Petr Pavel won with 3,359,301 votes, Andrej Babiš got just a little less, 2,399,898 votes. (Czech Statistical Office, 2023)

Correlation of vote shares and turnout and valid votes

Looking at the correlation coefficient values between the vote shares and turnout and valid votes in Figure 3.2, these near-zero values indicate no systematic relationship between turnout and candidate support. This suggests that voting patterns were not artificially concentrated in high-turnout municipalities.

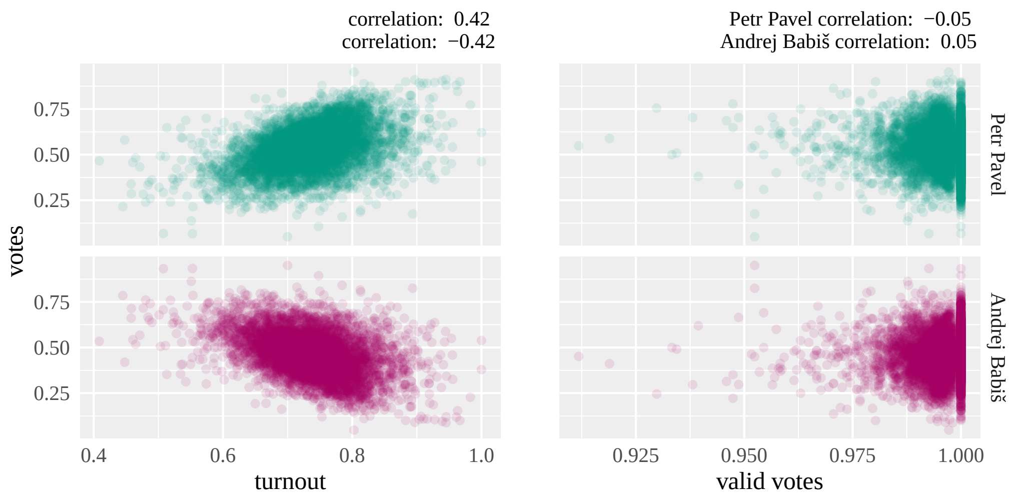
After dividing the plots between the two candidates in Figure 3.3, we can see that the correlations are the opposites for the candidates. This makes sense, since one of them is the winner, taking the vote share from the other one, and they were the only ones computing, so the sums must be equal to 1 of all the vote shares. In both graphs, we can see no obscure pattern that could indicate a fraud of any sort.

Figure 3.2: Correlation of vote share and turnout, and vote share and valid votes, all votes, Czechia 2023



Source: data: Czech Statistical Office (2023), processing: author

Figure 3.3: Correlation of vote share and turnout; and vote share and valid votes for each candidate, Czechia, 2023

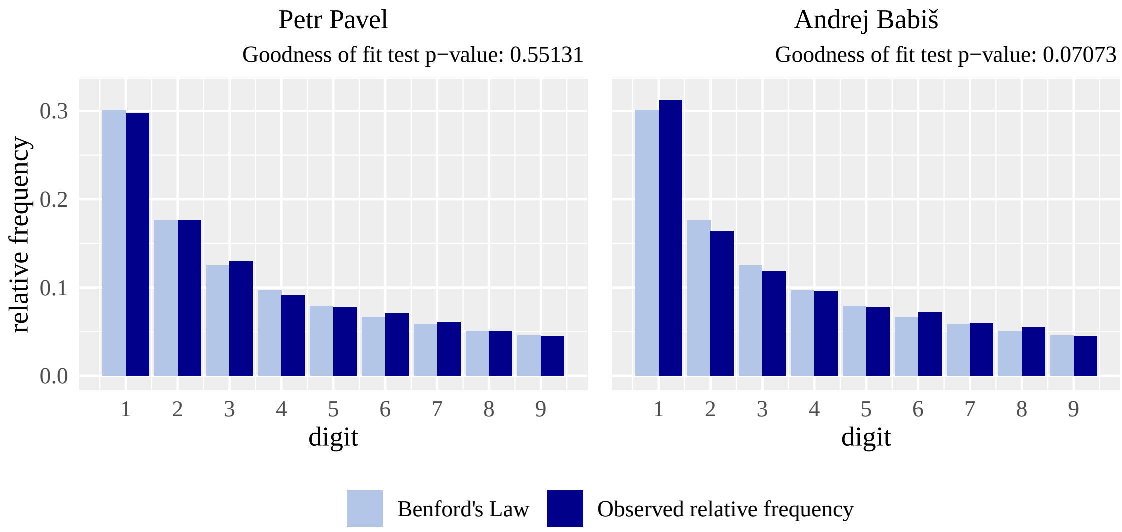


Source: data: Czech Statistical Office (2023), processing: author

Relative frequencies of the digits on a given position

In Figure 3.4, we can see how the data complies with Benford's first digit law. There are no visible large deviations of the observed relative frequencies from the expected theoretical frequencies. The goodness of fit tests support this by not rejecting their null hypothesis on the set significance level $\alpha = 0.05$. The null hypothesis suggests that the relative frequencies are equal. With the returned p-values, which are all above the α, we do not reject this null hypothesis for either candidate.

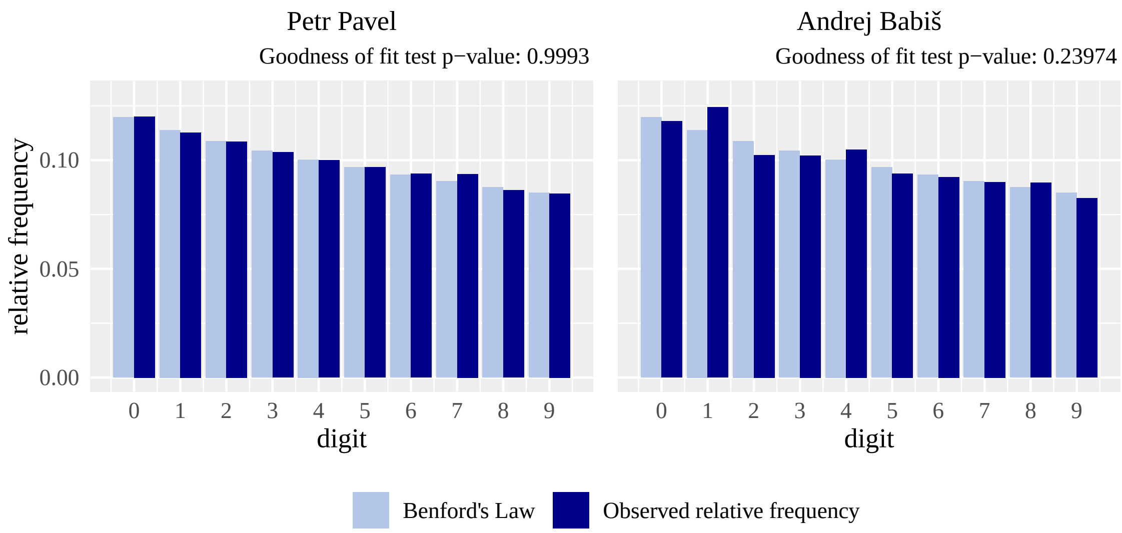
Figure 3.4: First digit distributions of both candidates, Czechia, 2023



Source: data: Czech Statistical Office (2023), processing: author

Then, the test for the second leading digits compliance is not rejected for either of the candidates, as can be seen in Figure 3.5. We can also witness an almost perfect compliance with the law of the votes for Petr Pavel, with very minimal deviations of the observed frequencies of the digits from the theoretical frequencies.

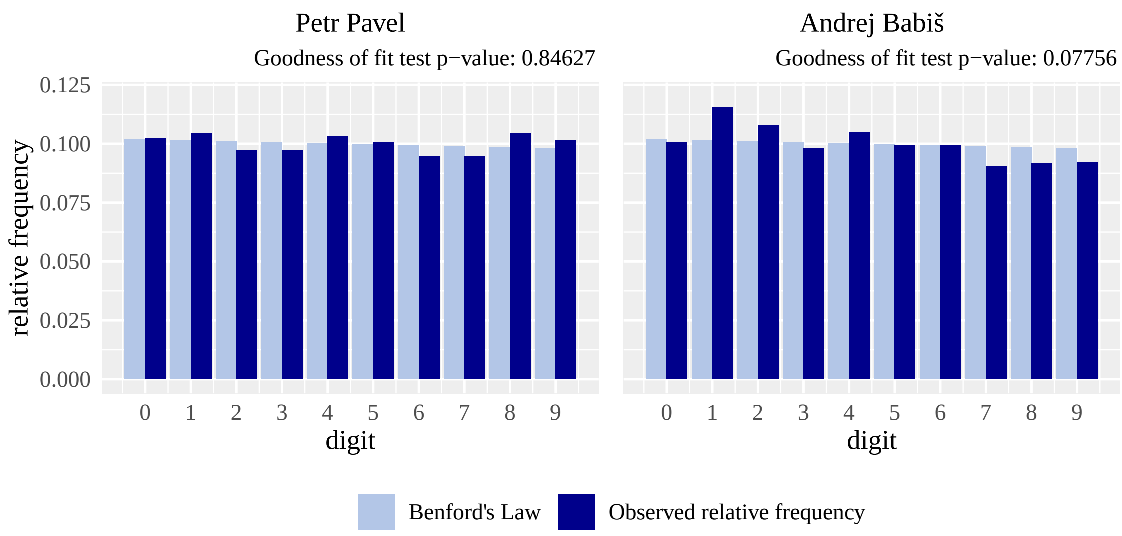
Figure 3.5: Second digit distributions from both candidates, Czechia, 2023



Source: data: Czech Statistical Office (2023), processing: author

Up next, the third digit distribution in Figure 3.6, again, does not provide any evidence of fraud. The observed higher frequency of the digit 1 in Babiš's votes is not enough to reject the null hypothesis at the stated significance level.

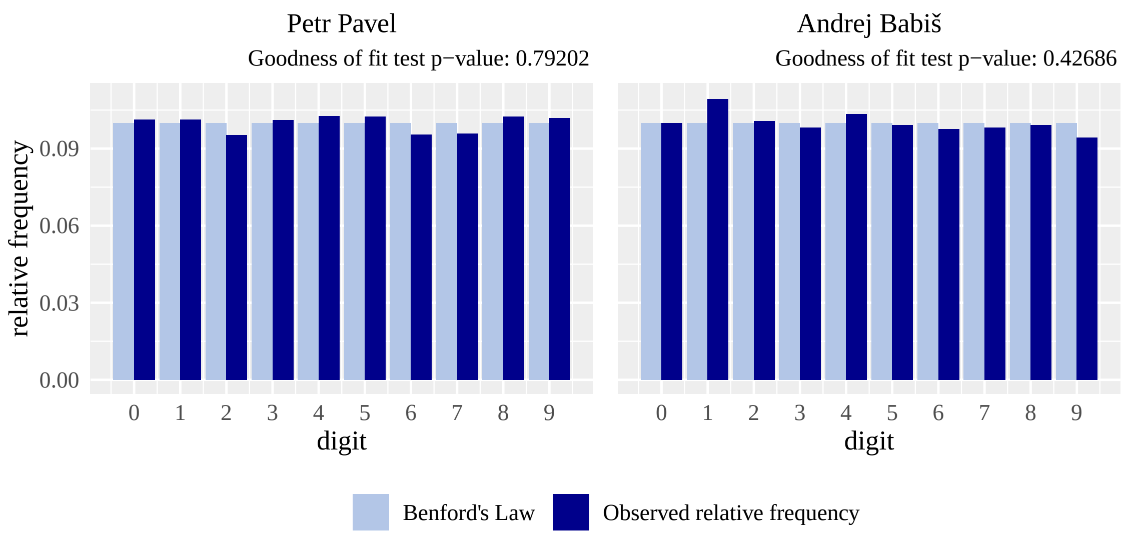
Figure 3.6: Third digits distributions from both candidates, Czechia, 2023



Source: data: Czech Statistical Office (2023), processing: author

And lastly, in Figure 3.7 we can see the compliance with the BL last digit law. The goodness of fit test does not reject the null hypothesis that the relative frequencies are close to equal. This does not provide evidence for rounding or any other manipulation of the vote counts.

Figure 3.7: Last digit distributions from both candidates, Czechia, 2023

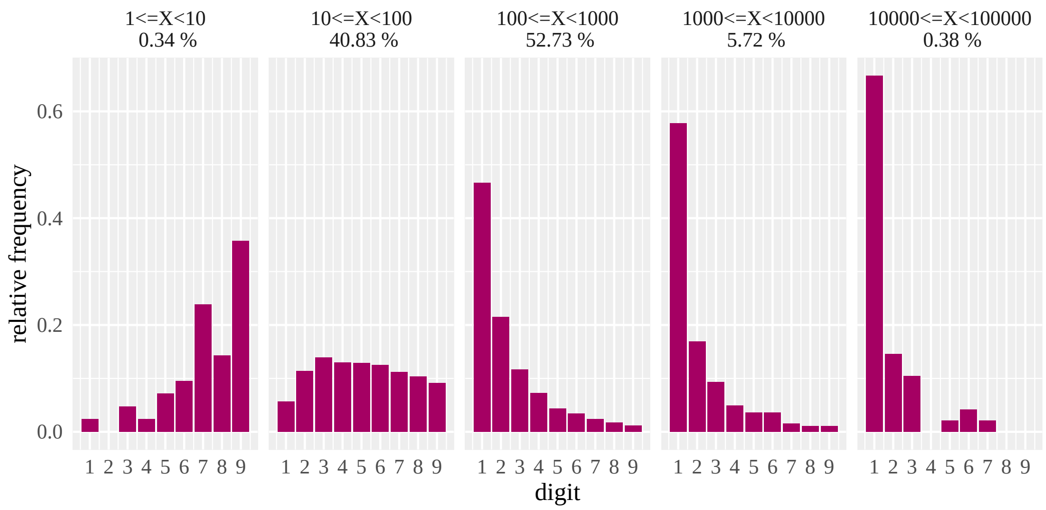


Source: data: Czech Statistical Office (2023), processing: author

Digital development pattern analysis

Observing the digital development pattern in Figure 3.8, which illustrates the first-digit frequency within five bins corresponding to order-of-magnitude ranges. Firstly, we can see that for the smallest integral orders of ten, which are represented by only 0.34% in the whole dataset, the higher they are, the more frequent they are. This aligns with what was expected, as described in 2.1.3 section. When going to the higher orders, the pattern inverts. In the most prevalent interval in the data (middle plot, 52.73%), the data is close to the BL first digit law. From there, the first digits are more frequent, as expected.
To conclude this part, the observed digital development pattern aligns with the expected pattern.

Figure 3.8: Digital development pattern, Czechia, 2023



Source: data: Czech Statistical Office (2023), processing: author

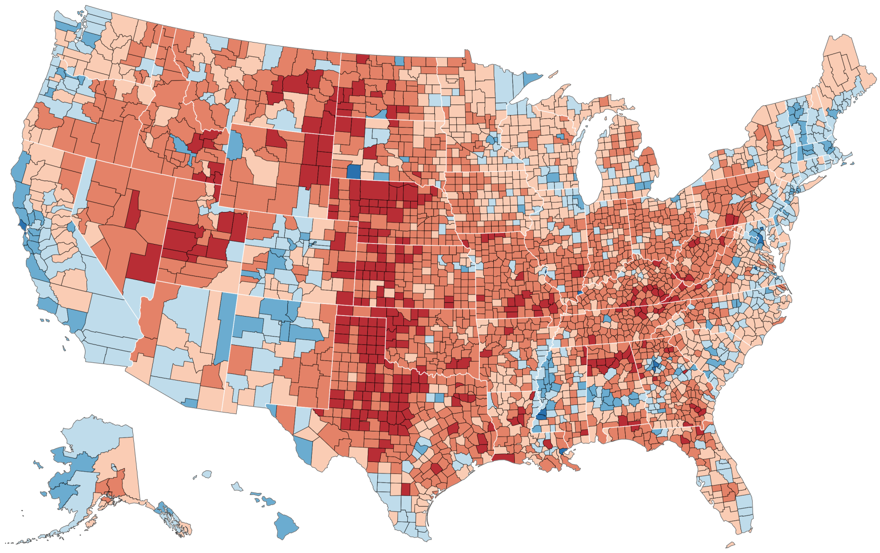
Results

In this analysis, we have not gained enough evidence to conclude fraud in this election. The correlation of vote shares and turnout is close to zero, as well as the correlation of the vote shares and the validity of votes. No speculative clusters were discovered. First digits comply with Benford's first digit law for the whole dataset and both candidates. The distribution of the second leading digits also did not show any deviations from the expected Benford's second digit law distribution, showing no traces of number rounding either. Neither did the third and the last digit distributions. Lastly, the digital development pattern showed no signs of digital irregularities. Overall, based on our results, **we can assume that the Czech presidential elections in 2023 are genuine and free of any digital fraud**. But that does not mean there may not be any fraud of another origin. This confirms the findings of the OSCE (2018) in the previous election, and to our knowledge, there is no suspicion in the media that these elections are fraudulent.

3.2 Presidential election in the USA in 2024

Following the analysis of the Czech presidential election, we turn our attention to the highly anticipated 2024 presidential election in the United States. This election garnered significant global attention due to some allegations of manipulation and fraud. (OBSE, 2025) Unlike Czechia's centrally published data, U.S. county-level returns must be aggregated from state websites; here we use the McGovern et al. (2025) compilation of 3,143 counties and county-equivalents scraped from Fox News.

Figure 3.9: The results of the presidential election in the USA in 2024



Source: McGovern et al., 2025

The election was held on November 5, 2024. The distribution of votes for the candidates can be seen in a map in Figure 3.9, where the blue shades represent a higher vote share for the democratic party representative, Kamala Harris and the red shades signify the higher vote shares of the republican, Donald Trump. The final results can be seen in Table 3.2. Donald Trump won the election with 77,302,580 votes. His main opponent, Kamala Harris, acquired 75,017,613 votes in total. (Federal Election Commission, 2025)

Table 3.2: Election results in the USA in 2024

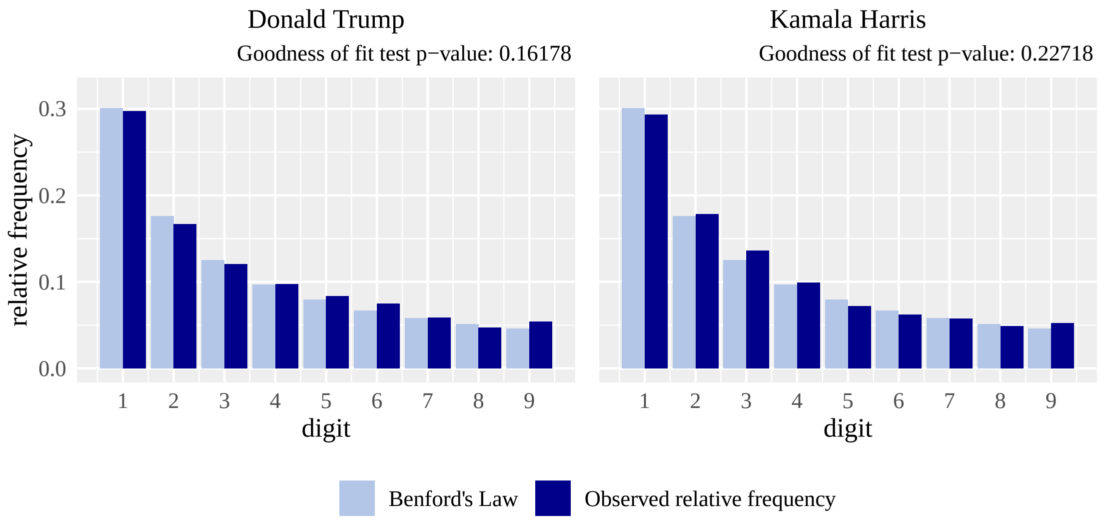
candidate	votes share
Donald Trump	49.8%
Kamala Harris	48.3%
other candidates	1.9%

Source: Federal Election Commission (2025)

Relative frequencies of the digits on a given position

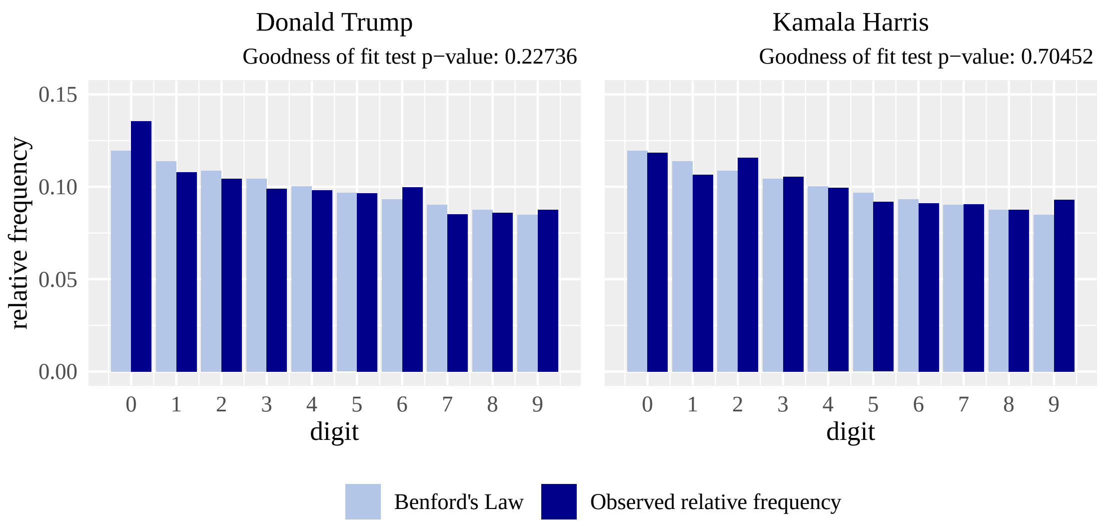
Unfortunately, the data about election turnout was not available, so the correlation of this variable and the vote shares cannot be assessed. The analysis nevertheless continues in this section with testing the compliance with the BL distributions. Looking at the first digit distribution for both candidates in Figure 3.10, we can see that the relative frequencies are pretty close to the BL probabilities. The goodness of fit tests are not rejected at the p-values higher than the level of significance $\alpha = 0.05$.

Figure 3.10: First digit distributions from both candidates, USA, 2024



Source: data: McGovern et al. (2025), processing: author

Figure 3.11: Second digit distributions from both candidates, USA, 2024

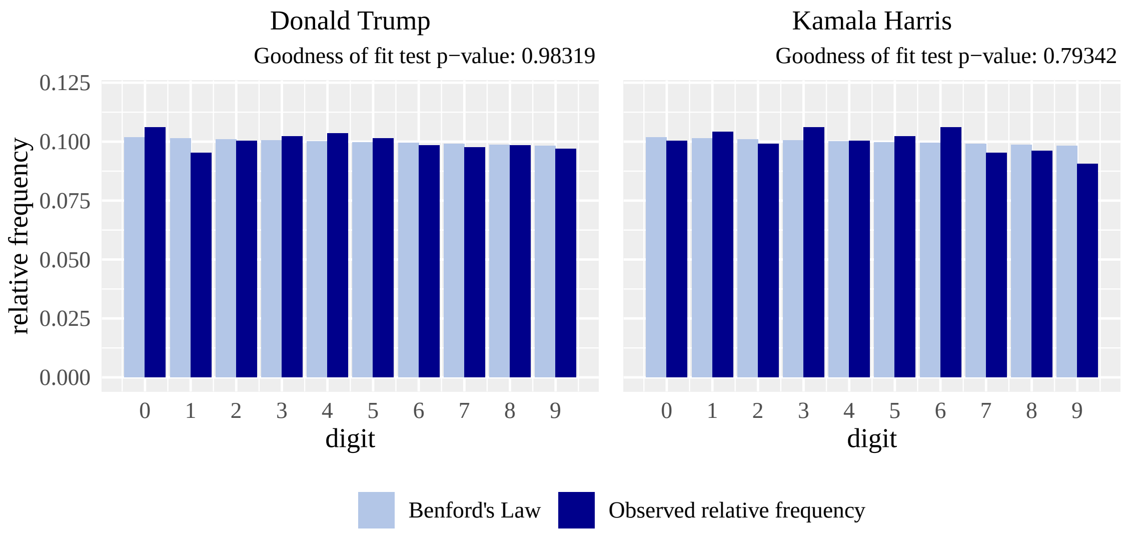


Source: data: McGovern et al. (2025), processing: author

In the second digit distribution, Trump's slight over-representation of digit 0 is well within random sampling variation ($p > 0.05$). There are no other deviations, and the goodness of fit tests are not rejected for either of the candidates, as can be seen in Figure 3.11.

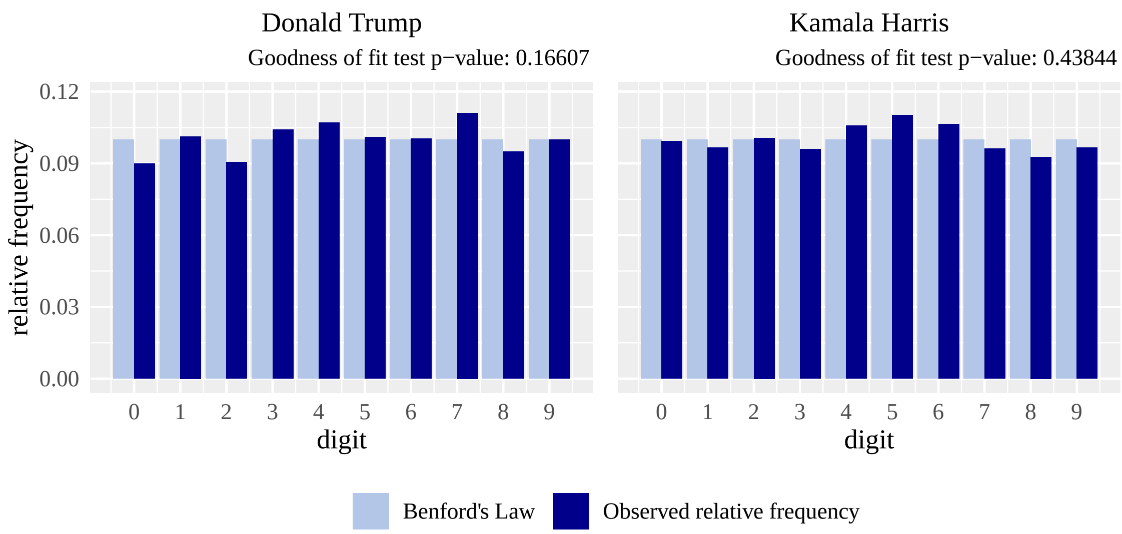
Moving on with the third leading digits distribution, in Figure 3.12 we can see that the observed distribution of relative frequencies of digits in the third position and the theoretical distribution according to Benford's Law comply with each other. This is supported by the p-values being larger than the set significance level $\alpha = 0.05$.

Figure 3.12: Third digits distributions from both candidates, USA, 2024



Source: data: McGovern et al. (2025), processing: author

Figure 3.13: Last digit distributions from both candidates, USA, 2024



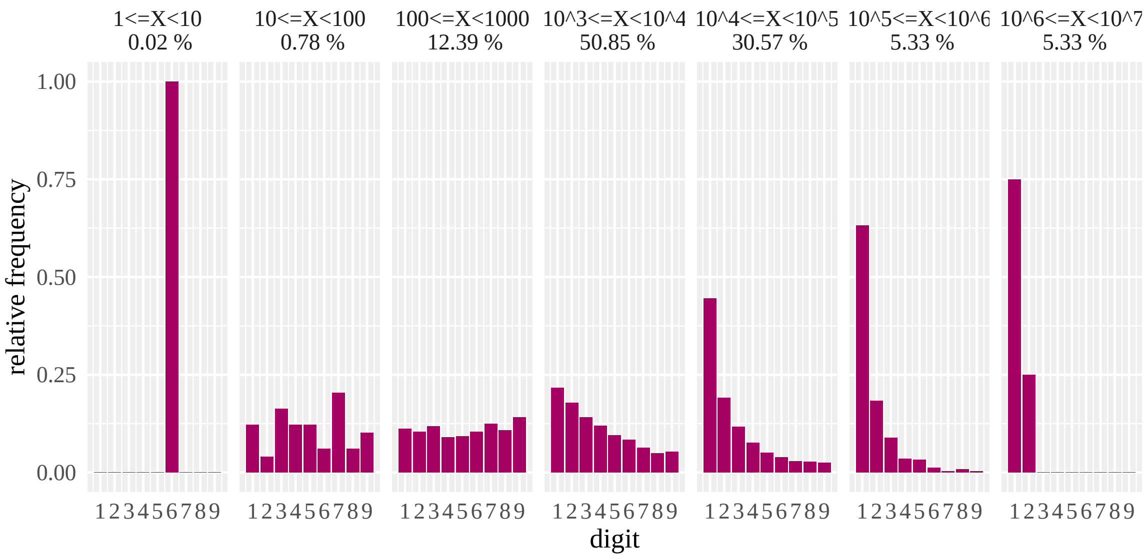
Source: data: McGovern et al. (2025), processing: author

Last digit distribution for both candidates in Figure 3.13 shows no signs of rounding numbers or any other deviations from the uniform distribution, and also the null hypothesis is not rejected.

Digital development pattern

In Figure 3.14, we have observed the development pattern of relative frequencies of the leading digits. In the lower values spectrum, the relative frequencies are not as skewed as we would expect; this is because the smallest bin is too sparse for a clear pattern, so the pattern does not have a chance to develop. In the rest of this bottom part, the distribution is closer to equal, with a slight rise towards Benford's first digit law. In the largest partition of the data, the relative frequencies are getting closer to the law. From there, the first digits are getting much more prevalent. This pattern is in alignment with what Kossovsky (2014) describes. We have reason to think otherwise.

Figure 3.14: Digital development pattern, USA, 2024



Source: data: McGovern et al. (2025), processing: author

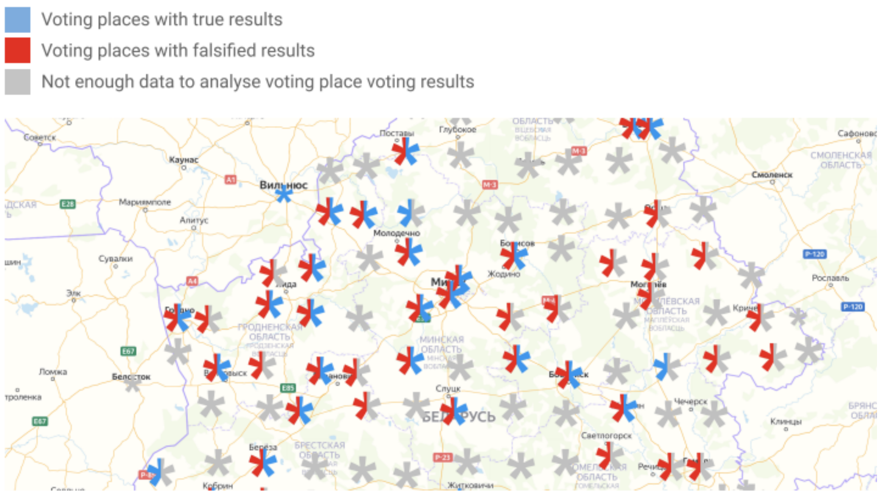
Results

In our tests, we have not acquired enough evidence to suspect fraud of the digital kind in the 2024 presidential election in the USA. The first significant digit distribution of the vote counts for both candidates complies with Benford's Law very well. The goodness of fit tests in the second leading digits distribution had not detected any deviations from Benford's Law either. Third, and then the last digit distribution had not detected any rounding of the vote counts to specific digits, and their p-values were higher than the set level of significance. The digital development pattern in these vote counts is in alignment with the expected behaviour. In conclusion, based on the results of our analysis, **we can assume that the USA presidential elections in 2024 are free of any digital fraud**. This aligns with the findings of ODHIR, the USA presidential election in 2024 "*did not indicate that the votes or election results were altered*" and that "*the claims of potential electoral fraud were unsubstantiated*". (OBSE, 2025)

3.3 Presidential elections in Belarus in 2020

The presidential elections were held on 9th August 2020. The process was labelled as unfair and not democratic by international organisations and monitors, and was quite controversial. The official election results, as seen in Table 3.3, have not been accepted to this day in the EU, the USA and other democratic societies and have been concluded fraudulent. (Bedford, 2021; Central Election Commission, 2020)

Figure 3.15: Polling stations with fraudulent and true results, Belarus, 2020



Source and processing: Voice in 2020

Their winner, Lukashenko, had been president of Belarus since 1994. His opponent, Tsikhanouskaya, had been supported by a large fraction of the Belarusian population. The election fraud was proven thanks to the Belarusian initiatives Zubr, Honest People and Voice by collecting registered votes from the people, which show that the results should have been very different; their work is visualised by a map, which can be seen in Figure 3.15. (Bedford, 2021; Honest People & Zubr, 2020; Voice, 2020)

Table 3.3: Reported election results in Belarus 2020

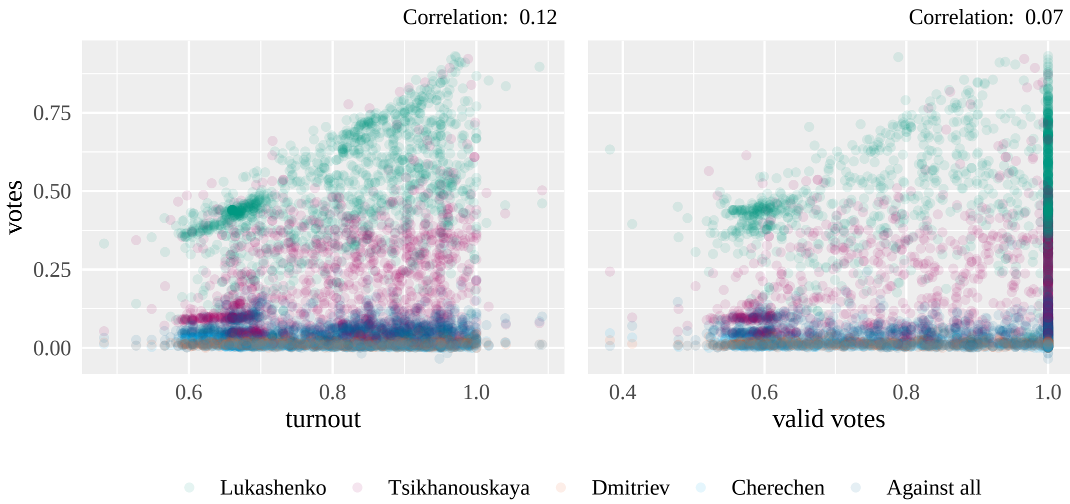
candidate	votes share
Alexander Lukashenko	81.04%
Sviatlana Tsikhanouskaya	10.23%
other candidates	8.73%

Source: Central Election Commission (2020)

Correlation of vote shares and turnout and valid votes

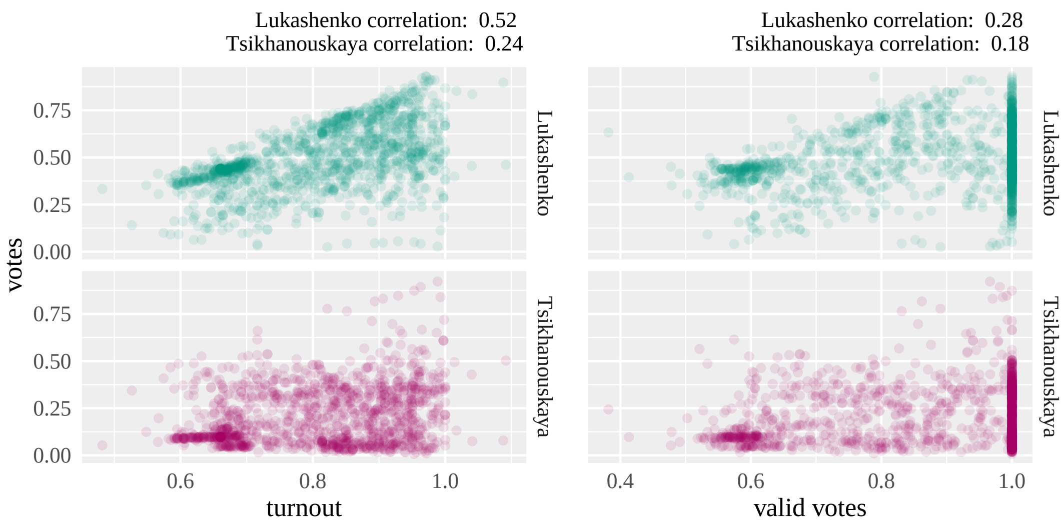
When examining the scatter plots in Figures 3.16 and 3.17, several notable observations emerge. Firstly, it is apparent that in some polling stations, the turnout exceeded 100%. This anomaly could be attributed to data entry errors or may indicate inconsistencies within the dataset, potentially signalling fraudulent activity.

Figure 3.16: Correlation of vote share and turnout; and vote and valid vote share, all candidates, Belarus, 2020



Source: data: Honest People and Zubr (2020), processing: author

Figure 3.17: Correlation of vote share and turnout; and vote share and valid votes, selected candidates, Belarus, 2020



Source: data: Honest People and Zubr (2020), processing: author

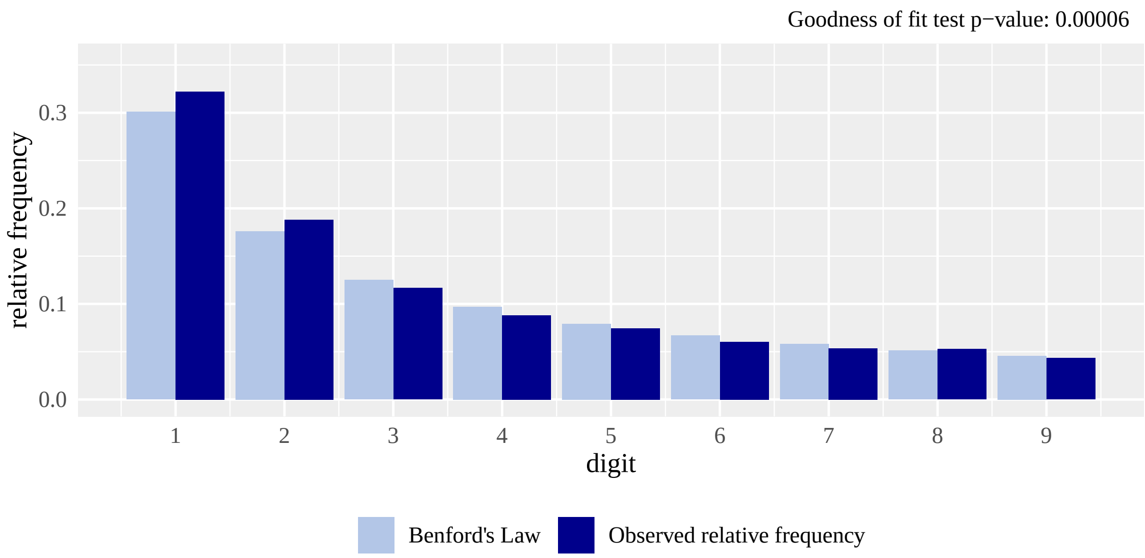
Additionally, the scatter plots reveal the presence of distinct clusters. Under normal circumstances, we would not expect clustering in these plots, except for the two *line-shaped* clusters, as explained in section 2.4.2. The existence of the other clusters suggests possible record manipulation, such as altering vote counts to achieve a desired vote share. This manipulation could be indicative of random efforts to skew the election results.

Moreover, the correlation coefficients provide further insight into the relationships between vote shares and turnout. For instance, the correlation between Lukashenko's vote share and turnout is notably high at 0.52, while Tsikhanouskaya's correlation is lower at 0.24, but not the opposite. This kind of pattern would be expected, as illustrated in the Czech presidential election (Figure 3.3), that one candidate's success means the loss for the other. Similarly, the correlation between valid votes and vote share for Lukashenko is 0.28, compared to 0.18 for Tsikhanouskaya. Additionally, we can see that the turnout is slightly dependent on the candidate's result among all the votes; the correlation coefficient is more than 0.1.

Relative frequencies of the digits on a given position

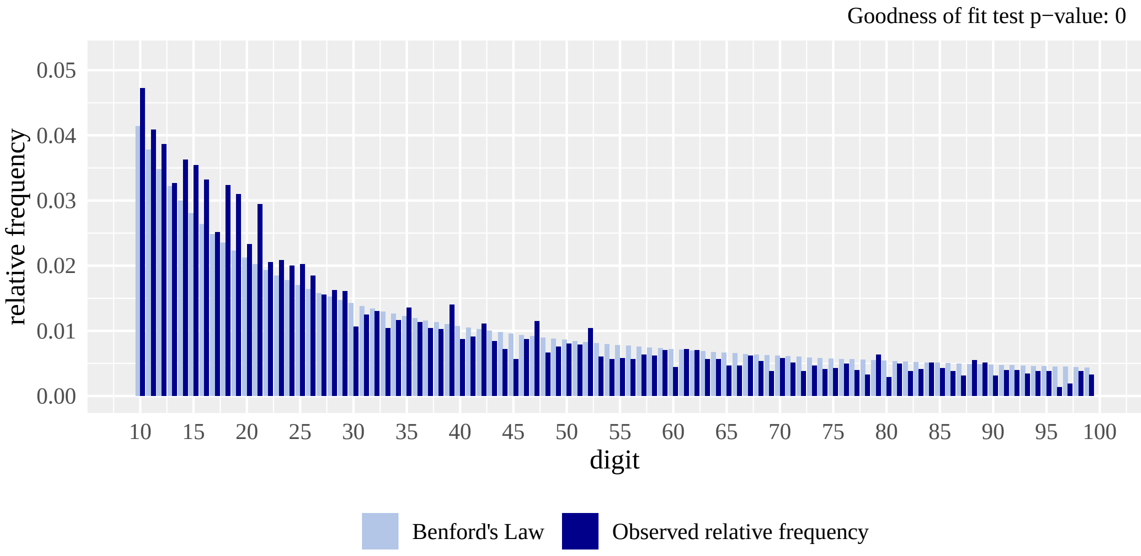
When visually inspecting the Figure 3.18, which shows the first digit distribution, the plot does look like the data complies with Benford's first leading digit law. However, the p-value from the goodness of fit test is close to zero, lower than the level of significance as set in the previous sections in this thesis; thus, **the null hypothesis**, which formulated that the relative frequencies are equal, **is rejected**. For a more detailed view of the first digits distribution, look at the distribution of the first two digits in Figure 3.19, which also rejects the null hypothesis and shows the deviances in greater detail.

Figure 3.18: First digit distribution, Belarus, 2020



Source: data: Honest People and Zubr (2020), processing: author

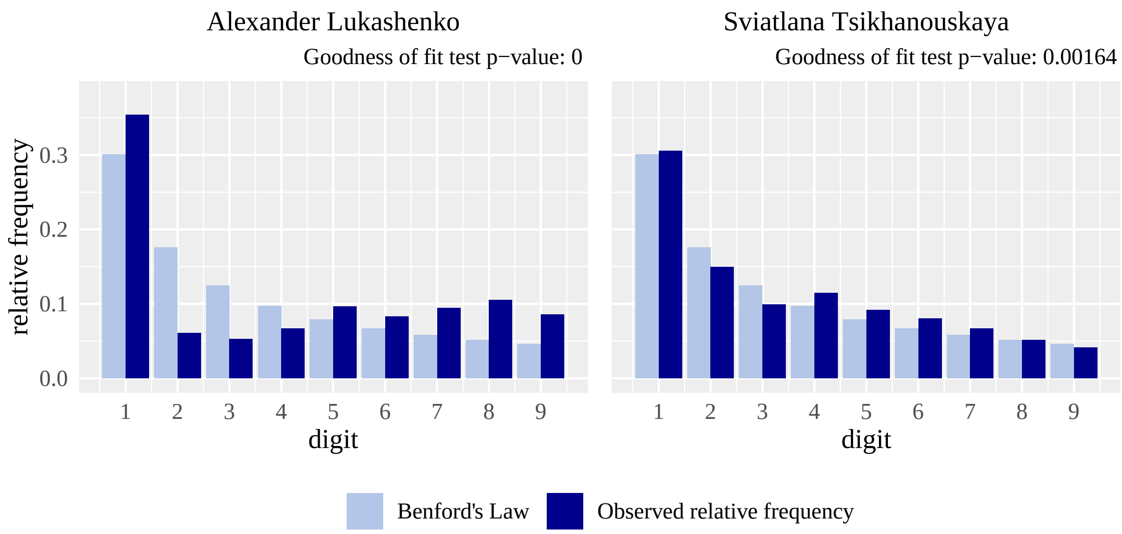
Figure 3.19: First two digits distribution, Belarus, 2020



Source: data: Honest People and Zubr (2020), processing: author

All becomes apparent when the two rivals' relative frequencies of digits in their vote counts are plotted separately in Figure 3.20. For both candidates, the tests provide evidence of fraudulent results. The pattern for Tsikhanouskaya follows the BL more closely, the goodness of fit test yields a p-value of 0.00164, than for Lukashenka, with the p-value of very close to 0, indicating a notable deviation from Benford's Law, while the digit distribution of the vote counts for him is very unusual. This provides very strong evidence that the records may be manipulated.

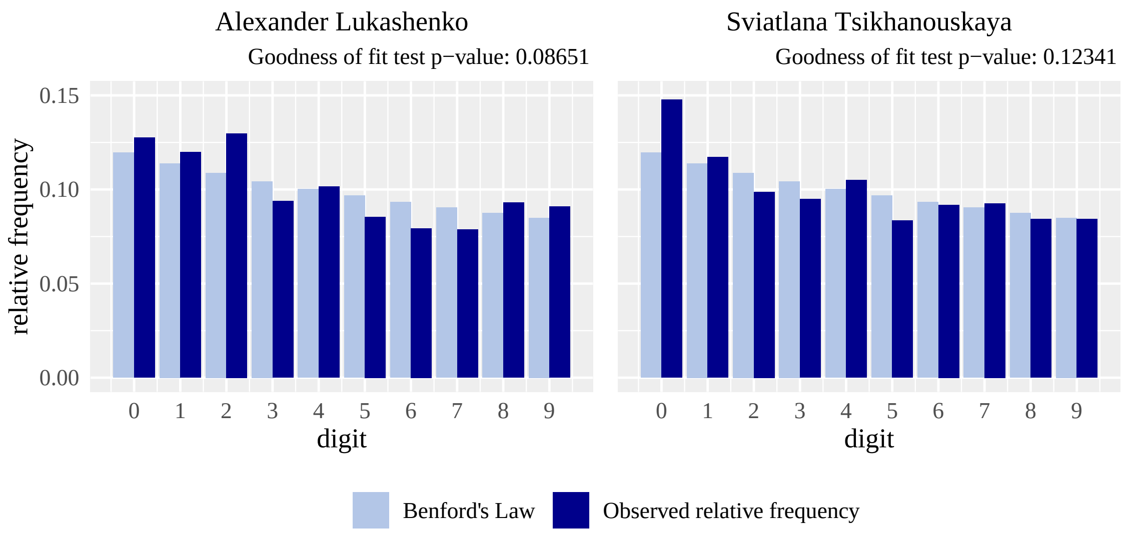
Figure 3.20: First digit distributions from both candidates, Belarus, 2020



Source: data: Honest People and Zubr (2020), processing: author

In contrast to the strong deviations observed in the first-digit distributions (Figure 3.20), the second-digit distributions for both candidates in Figure 3.21 exhibit a close alignment with Benford's Law. This is supported by the results of the chi-squared goodness-of-fit tests, which yield p-values for both candidates comfortably above the significance level.

Figure 3.21: Second digit distributions from both candidates, Belarus, 2020



Source: data: Honest People and Zubr (2020), processing: author

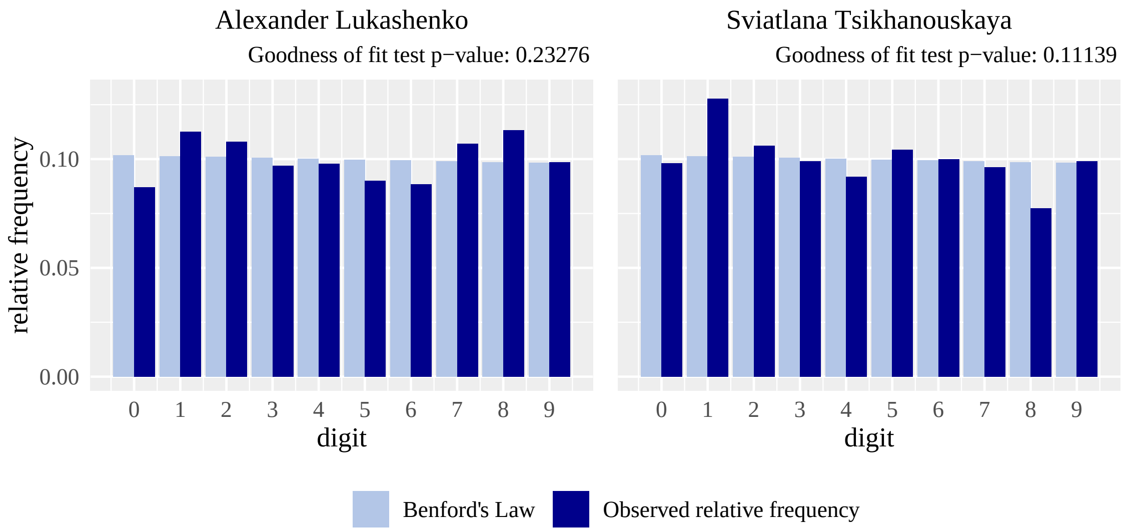
The observed frequencies fluctuate mildly around Benford's expected values, but without the pronounced distortions seen in manipulated datasets. This suggests that, at least at the level of second digits, the vote totals do not exhibit strong evidence of artificial rounding or mechanical fabrication. However, it is important to emphasise that compliance with Benford's Law at one digit level does not imply the absence of manipulation overall. Malpractice may be more detectable in the first-digit or last-digit patterns, or may manifest in other structural anomalies not captured by digit analysis alone.

Continuing the analysis, we turn to the third and last digit distributions, shown in Figures 3.22 and 3.23. In Figure 3.22, the third-digit frequencies for both candidates exhibit a reasonably close fit to Benford's Law. The χ^2 goodness-of-fit test yields p-values of 0.23276 for Lukashenko and 0.1139 for Tsikhanouskaya, both well above the 0.05 significance threshold. This indicates that, statistically, the third-digit distribution does not deviate significantly from the expected pattern.

In contrast, Figure 3.23 reveals greater deviation in the last-digit distributions. Lukashenko's data in particular shows a low p-value of 0.01839, falling below the 0.05 threshold and suggesting non-conformity with Benford's distribution. Tsikhanouskaya's last-digit distribution is borderline, with a p-value slightly above the threshold.

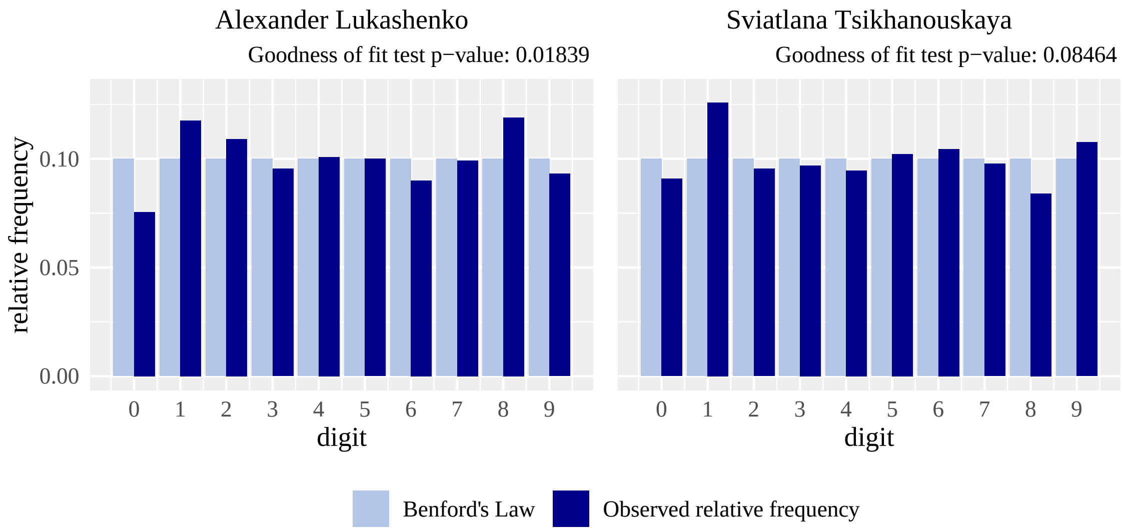
The stronger deviation in the last-digit distribution, especially for Lukashenko, may point to unnatural rounding or other manipulation. Since the last digits are typically expected

Figure 3.22: Third digits distributions from all candidates, Belarus, 2020



Source: data: Honest People and Zubr (2020), processing: author

Figure 3.23: Last digit distributions from both candidates, Belarus, 2020



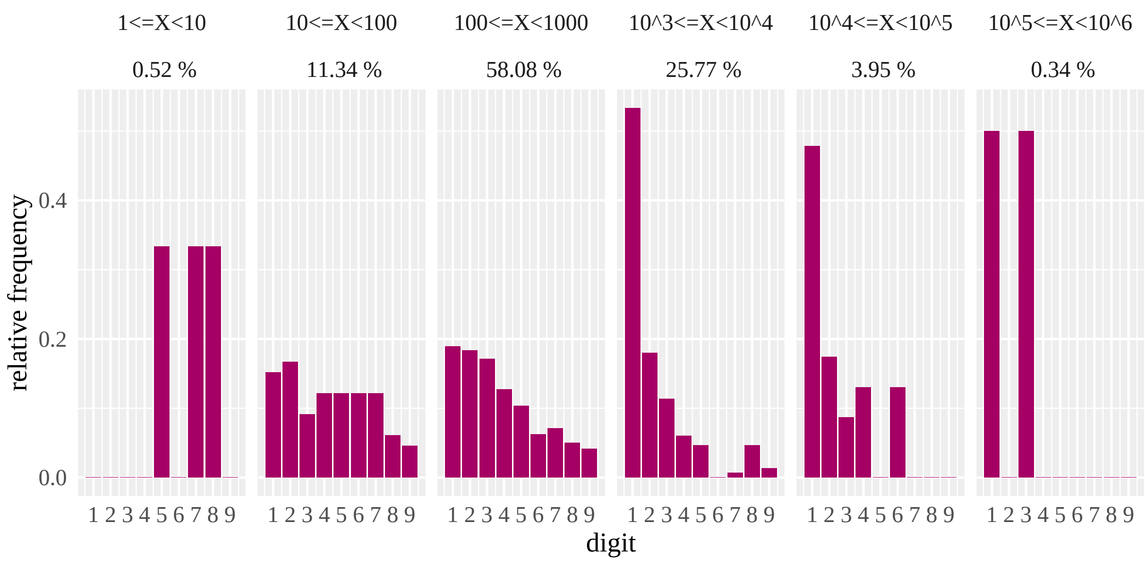
Source: data: Honest People and Zubr (2020), processing: author

to follow a uniform distribution as described by Hill (1995), their distortion is often regarded as a red flag in election forensics. The third-digit results, by contrast, show no such anomalies. The differing outcomes between candidates further support the hypothesis that Lukashenko's reported results were subject to greater manipulation or irregularities than those of Tsikhanouskaya.

Digital development pattern

Observing the digital development pattern in the original data on the individual polling station level proved infeasible due to the structure of the dataset: most polling stations had a very similar number of eligible voters, resulting in limited variation in vote counts. This homogeneity prevents the emergence of a natural logarithmic distribution across orders of magnitude. However, when the data is aggregated to the city level, as shown in Figure 3.24, it becomes possible to assess the development of digit frequencies across bins of increasing magnitude.

Figure 3.24: Digital development pattern - aggregated data on a city level, Belarus, 2020



Source: data: Honest People and Zubr (2020), processing: author

In this aggregated form, several irregularities become apparent. While the first and last bins should be disregarded due to a low number of observations, the intermediate bins display deviations from expected behaviour. Notably, the second bin does not exhibit the expected equalisation of digit frequencies, and higher-magnitude bins fail to show the anticipated convergence toward lower digits (particularly digit 1) dominating. This absence of the expected trend, where lower digits become more frequent as magnitude increases, suggests that the aggregated vote totals may reflect some unnatural patterns.

Results

In democratic societies, the election results are publicly available in detail. This is not the case in Belarus, where we had to use just a sample of election protocols from one-fifth of all polling stations and only thanks to democratic initiatives like Voice, Zubr and the Honest People. This does not put a good light on the election integrity in Belarus.

Several findings emerged from the analysis of the 2020 presidential election in Belarus. There is a slight correlation between a candidate's vote share, notably Lukashenko's, and turnout. Turnout at some polling stations exceeded 100%. Some unexpected clusters of polling stations of potentially strange origin were also discovered. All of which could suggest dubious activity.

Regarding the digit distributions, the analysis reveals notable irregularities, particularly in the vote counts attributed to Alexander Lukashenko. The first significant digit distribution of the vote counts for both candidates does not comply with Benford's law. This is further enhanced by observing the distribution of the first digits for each candidate. Lukashenko's results deviate significantly from Benford's Law, with a p-value close to zero and by a very visually different distribution, raising suspicion of data manipulation. This is supported by the rejection of Lukashenko's last digit distribution. Tsikhanouskaya's last-digit p-value is borderline, but still not indicative of significant deviation.

In contrast, the second and third digit distributions align more closely with the expected pattern, implying fewer anomalies in her reported vote totals, indicating that any manipulation, if present, may not have been due to rounding up the digits, but rather to some random changes in the actual vote counts. However, the hypothesis of the existing unnatural patterns in data is supported by visual inspection of the digital development pattern. The frequencies of the first digits do not behave as expected.

Taken together, these results support the broader hypothesis echoed by election observers and civic groups: **that the vote counts were likely manipulated**. Especially the vote counts attributed to Lukashenko, these may have been changed to a higher degree than those of his opponent, particularly in ways that distort the natural structure of the data. This finding is in alignment with the opinion of the EU and other democratic countries and the concern expressed by the ODHIR, as these international observers of elections had not been invited. (Council of the EU, 2020; OBSE, 2020a)

Conclusion

This thesis presented Benford's Law in the context of uncovering election fraud. To explore the applicability of Benford's Law for detecting irregularities in election results, the method was applied to presidential election data. The theoretical part introduced the historical background and mathematical formulation of the law, along with the conditions under which it can be validly applied. Elections and electoral integrity were described. The statistical methods and testing strategies used were also introduced and implemented. By the last page, **all three aims of this thesis were fulfilled**.

This thesis provides an introduction to Benford's Law for election fraud detection in its first chapter. Including its historical background, mathematical formulation, and the conditions under which it is applicable, along with the description of the election process, the election integrity and addressing what election fraud is. Thus, **the first aim is fulfilled**.

To contribute to the protection of electoral integrity, the author designed a reproducible methodology for detecting potential digital anomalies in vote counts. This methodology applies Benford's Law as an underlying distribution of digits in a given position. The compliance is tested by the goodness of fit test. And it is accompanied by other indicative measures, such as the correlation of relevant variables and the development of the digital pattern. All wrapped in a structured and clear analytical workflow, freely available in an accessible form at GitHub (Bednářová, 2025). and thus **the second aim is fulfilled**.

The author's proposed methodology was applied to the results of the presidential election in the Czech Republic in 2023, in the USA in 2024, and in Belarus in 2020. The results of the analyses were compared to the relevant institutions' and international organisations' findings to draw meaningful and evidence-based conclusions. Although more comprehensive and complex approaches to election forensics exist, the author demonstrates that even a relatively simple methodology, when carefully applied, can yield conclusions that align with those of respected institutions and independent observers. And thus, **the third aim is fulfilled**.

The analysis showed that the election results from the Czech Republic and the United States align closely with the expected Benford distributions, particularly for the first and second digits. No statistically significant deviations were found in these cases that would suggest manipulation. this is supported by the reports by OBSE (2018, 2023, 2025). In contrast, the results from Belarus revealed substantial deviations, especially in the results for candidate Lukashenko, suggesting the manipulation of a higher degree than that of his opponent. While this cannot serve as conclusive evidence of fraud, it strongly suggests inconsistencies in the data. This is supported by the opinion of the Council of the EU (2020) and other democratic countries and the concern expressed by uninvited observers of the election, OBSE (2020a, 2020b).

Benford's Law proved to be a useful tool for preliminary statistical uncovering of election fraud by identifying the presence of artificially constructed numbers. However, it is not a universal fraud detector and should be combined with other methods, as described in Lebeda et al. (2021). Its limitations include the inability to detect other forms of electoral misconduct, such as voter intimidation, ballot suppression, or other types of election fraud as described by Levitt (2007) or Lehoucq (2003). Future research may expand this work by comparing election years within a country or integrating Benford's Law into machine learning models for anomaly detection.

This thesis contributes to the ongoing discussion about the role of statistical tools in electoral forensics and illustrates both the potential and the boundaries of Benford's Law in the context of evaluating electoral data.

List of references

Beber, B., & Scacco, A. (2012). What the Numbers Say: A Digit-Based Test for Election Fraud. *Political Analysis*, *20*(2), 211–234. https://doi.org/10.1093/pan/mps003

Bedford, S. (2021). The 2020 Presidential Election in Belarus: Erosion of Authoritarian Stability and Re-politicization of Society. *Nationalities Papers*, *49*(5), 808–819. https://doi.org/10.1017/nps.2021.33

Bednářová, K. (2025). Benford's law and the statistical detection of election fraud [GitHub repository]. https://github.com/Kikibedna/benfords-law-election-fraud

Benford, F. (1938). The Law of Anomalous Numbers. *Proceedings of the American Philosophical Society*, *78*(4), 551–572. Retrieved 2024-09-06, from http://www.jstor.org/stable/984802

Bouchal, P. (2024). *Czso: Use open data from the Czech Statistical Office in R* [R package version 0.4.1]. https://CRAN.R-project.org/package=czso

Central Election Commission. (2020). *Voting results*. Retrieved 2025-05-10, from https://web.archive.org/web/20200929235004/http://rec.gov.by/sites/default/files/pdf/2020/inf9.pdf

Cerqueti, R., & Lupi, C. (2022). Severe Testing of Benford's Law. *TEST*, *32*, 677–694. https://api.semanticscholar.org/CorpusID:246706184

Council of the EU. (2020). *Belarus: Eu imposes sanctions for repression and election falsification* [press release]. Retrieved 2025-05-10, from https://www.consilium.europa.eu/en/press/press-releases/2020/10/02/belarus-eu-imposes-sanctions-for-repression-and-election-falsification/pdf/

Czech Statistical Office. (2023). *Election of the President of the Czech Republic held on 13–14 January 2023*. Retrieved 2025-04-18, from https://www.volby.cz/pls/prez2023nss/pe71?xjazyk=EN#/0-01-2023-0

Deckert, J., Myagkov, M., & Ordeshook, P. C. (2011). Benford's Law and the Detection of Election Fraud. *Political Analysis*, *19*(3), 245–268. https://doi.org/10.1093/pan/mpr014

Eulau, H., Webb, P., & Gibbins, R. (2025). Election. *Encyclopedia Britannica*. https://www.britannica.com/topic/election-political-science

Federal Election Commission. (2025). *Official 2024 Presidential General Election Results*. Retrieved 2025-04-18, from https://www.fec.gov/resources/cms-content/documents/2024presgeresults.pdf

Heumann, C., Schomaker, M., & Shalabh. (2022). *Introduction to statistics and data analysis: With exercises, solutions and applications in R*. Springer International Publishing. https://doi.org/10.1007/978-3-031-11833-3

Hill, T. P. (1995). A statistical derivation of the significant-digit law. *Statistical Science*, *10*(4), 354–363. Retrieved 2025-05-06, from http://www.jstor.org/stable/2246134

Honest People & Zubr. (2020). *2020 Belarusian presidential election - Processed data and photos of protocols of precinct election commissions*. Retrieved 2025-04-18, from https://www.kaggle.com/datasets/vadimkonovod/2020-belarusian-presidential-election/data

Hronová, S., Hindls, R., & Marek, L. (2023). Can Benford's Law Reflect Major Economic Changes? https://doi.org/10.5281/ZENODO.8060093

Kossovsky, A. (2014). *Benford's Law: Theory, The General Law of Relative Quantities, and Forensic Fraud Detection Applications*. World Scientific Publishing Company.

Lebeda, T., Šaradín, P., Lebedová, E., Kokeš, M., & Lysek, J. (2021-10). Volební pochybení a podvody ve volebních místnostech. Jak přispět k vyšší integritě voleb v České republice? *Acta Politologica*, *13*(3), 24–45. https://doi.org/10.14712/1803-8220/19_2021

Lehoucq, F. (2003-06). Electoral fraud: Causes, Types, and Consequences. *Annual Review of Political Science*, *6*(1), 233–256. https://doi.org/10.1146/annurev.polisci.6.121901.085655

Levitt, J. (2007). The Truth About Voter Fraud. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.1647224

Malá, I. (2024). *Statistické úsudky* (2nd ed.). Professional Publishing s.r.o.

Marek, L. (2024). *Pravděpodobnost* (Druhé vydání). Professional Publishing.

McGovern, T., Larson, S., Morris, B., & Hodges, M. (2025-01). *United States General Election Presidential Results by District, Ward, or County from 2008 to 2024*. Zenodo. https://doi.org/10.5281/zenodo.14223604

Mebane, W. R. (2011). Comment on "Benford's Law and the Detection of Election Fraud". *Political Analysis*, *19*(3), 269–272. https://doi.org/10.1093/pan/mpr024

Nejvyšší správní soud ČR. (2023). *Volební senát rozhodl všechny stížnosti na volbu prezidenta republiky* [press release]. Retrieved 2025-05-10, from https://www.nssoud.cz/aktualne/tiskove-zpravy/detail/volebni-senat-rozhodl-vsechny-stiznosti-na-volbu-prezidenta-republiky

Newcomb, S. (1881). Note on the Fequency of Use of the Different Digits in Natural Numbers. *American Journal of Mathematics*, *4*(1/4), 39. https://doi.org/10.2307/2369148

OBSE. (2018). *The Czech Republic parliamentary elections 20-21 October 2017 presidential election January 2018 OSCE/ODIHR needs assessment mission report*. Retrieved 2025-05-01, from https://www.osce.org/files/f/documents/9/0/333691.pdf

OBSE. (2020a). *ODIHR gravely concerned at situation in Belarus following presidential election*. Retrieved 2025-05-01, from https://www.osce.org/odihr/belarus/459664

OBSE. (2020b). *ODIHR will not deploy election observation mission to Belarus due to lack of invitation*. Retrieved 2025-05-01, from https://www.osce.org/odihr/belarus/457309

OBSE. (2023). *The Czech Republic presidential election 13-14 and 27-28 January 2023 ODIHR election expert team final report*. Retrieved 2025-05-01, from https://www.osce.org/files/f/documents/2/0/545815.pdf

OBSE. (2025). *United States of America general elections 5 November 2024 ODIHR limited election observation mission final report*. Retrieved 2025-05-01, from https://www.osce.org/files/f/documents/d/e/588772.pdf

Pedersen, T. L. (2024). *Patchwork: The composer of plots* [R package version 1.3.0]. https://CRAN.R-project.org/package=patchwork

R Core Team. (2023). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/

The Heritage Foundation. (2024-02). Categories of Election Fraud. https://electionfraud.heritage.org/categories

USA.gov. (2025). *Electoral college*. Retrieved 2025-05-01, from https://www.usa.gov/electoral-college

Voice. (2020). *Voting results*. Retrieved 2025-05-01, from https://partizan-results.com/?lon=27.543971&lat=51.724501&z=6

Wickham, H. (2016). *Ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. https://ggplot2.tidyverse.org

Wickham, H. (2023). *Stringr: Simple, consistent wrappers for common string operations* [R package version 1.5.1]. https://CRAN.R-project.org/package=stringr

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). *Dplyr: A grammar of data manipulation* [R package version 1.1.4]. https://CRAN.R-project.org/package=dplyr

Wickham, H., Hester, J., & Bryan, J. (2024). *Readr: Read rectangular text data* [R package version 2.1.5]. https://CRAN.R-project.org/package=readr

Wickham, H., Vaughan, D., & Girlich, M. (2024). *Tidyr: Tidy messy data* [R package version 1.3.1]. https://CRAN.R-project.org/package=tidyr

Zákon č. 88/2024 Sb. o správě voleb, In: *Zákony pro lidi.cz* [online]. © AION CS 2010–2025 (2024). https://www.zakonyprolidi.cz/cs/2024-88